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### Prosthetic valve having a multi-part frame

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#### Abstract

A prosthetic heart valve includes a multi-part frame having a first annular body and an anchor component that includes a second annular body circumscribing the first annular body. The first annular body includes a plurality of struts arranged in a lattice pattern and defining a lumen. The anchor component further includes one or more ventricular anchors extending from the second annular body toward an inflow end portion of the first annular body. A fabric layer is disposed on an inner surface of the first annular body and sutured to the struts. A valve structure including three leaflets is disposed within the lumen and inlet ends of the leaflets are sutured to the fabric layer. The multi-part frame is radially compressible for delivery within a sheath of a delivery catheter and is self-expandable for deployment within an annulus of a native valve.

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| 7018406 | 12/2005 | Seguin et al.        | N/A      | N/A         |
| 7018408 | 12/2005 | Bailey et al.        | N/A      | N/A         |
| 7048754 | 12/2005 | Martin et al.        | N/A      | N/A         |
| 7101395 | 12/2005 | Tremulis et al.      | N/A      | N/A         |
| 7125421 | 12/2005 | Tremulis et al.      | N/A      | N/A         |
| 7153322 | 12/2005 | Alt                  | N/A      | N/A         |
| 7186265 | 12/2006 | Sharkawy et al.      | N/A      | N/A         |
| 7192440 | 12/2006 | Andreas et al.       | N/A      | N/A         |
| 7198646 | 12/2006 | Figulla et al.       | N/A      | N/A         |
| 7201772 | 12/2006 | Schwammenthal et al. | N/A      | N/A         |
| 7261732 | 12/2006 | Justino              | 623/2.15 | A61F 2/2412 |
| 7276078 | 12/2006 | Spenser et al.       | N/A      | N/A         |
| 7276084 | 12/2006 | Yang et al.          | N/A      | N/A         |
| 7288097 | 12/2006 | Seguin               | N/A      | N/A         |
| 7318278 | 12/2007 | Zhang et al.         | N/A      | N/A         |
| 7329278 | 12/2007 | Seguin et al.        | N/A      | N/A         |
| 7371210 | 12/2007 | Brock et al.         | N/A      | N/A         |
| 7374571 | 12/2007 | Pease et al.         | N/A      | N/A         |

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| 7381210 | 12/2007 | Zarbatany et al.     | N/A      | N/A         |
| 7381219 | 12/2007 | Salahieh et al.      | N/A      | N/A         |
| 7393360 | 12/2007 | Spenser et al.       | N/A      | N/A         |
| 7429269 | 12/2007 | Schwammenthal et al. | N/A      | N/A         |
| 7442204 | 12/2007 | Schwammenthal et al. | N/A      | N/A         |
| 7445631 | 12/2007 | Salahieh et al.      | N/A      | N/A         |
| 7462191 | 12/2007 | Spenser et al.       | N/A      | N/A         |
| 7464712 | 12/2007 | Oz et al.            | N/A      | N/A         |
| 7509959 | 12/2008 | Oz et al.            | N/A      | N/A         |
| 7510575 | 12/2008 | Spenser et al.       | N/A      | N/A         |
| 7524330 | 12/2008 | Berrekouw            | N/A      | N/A         |
| 7530253 | 12/2008 | Spenser et al.       | N/A      | N/A         |
| 7553324 | 12/2008 | Andreas et al.       | N/A      | N/A         |
| 7569062 | 12/2008 | Kuehn et al.         | N/A      | N/A         |
| 7569071 | 12/2008 | Haverkost et al.     | N/A      | N/A         |
| 7579381 | 12/2008 | Dove                 | N/A      | N/A         |
| 7585321 | 12/2008 | Cribier              | N/A      | N/A         |
| 7618446 | 12/2008 | Andersen et al.      | N/A      | N/A         |
| 7618449 | 12/2008 | Tremulis et al.      | N/A      | N/A         |
| 7621948 | 12/2008 | Herrmann et al.      | N/A      | N/A         |
| 7682369 | 12/2009 | Seguin               | N/A      | N/A         |
| 7704222 | 12/2009 | Wilk et al.          | N/A      | N/A         |
| 7704277 | 12/2009 | Zakay et al.         | N/A      | N/A         |
| 7731706 | 12/2009 | Potter               | N/A      | N/A         |
| 7736327 | 12/2009 | Wilk et al.          | N/A      | N/A         |
| 7744609 | 12/2009 | Allen et al.         | N/A      | N/A         |
| 7748389 | 12/2009 | Salahieh et al.      | N/A      | N/A         |
| 7753932 | 12/2009 | Gingrich et al.      | N/A      | N/A         |
| 7753949 | 12/2009 | Lamphere et al.      | N/A      | N/A         |
| 7758596 | 12/2009 | Oz et al.            | N/A      | N/A         |
| 7780723 | 12/2009 | Taylor               | N/A      | N/A         |
| 7785360 | 12/2009 | Freitag              | N/A      | N/A         |
| 7803185 | 12/2009 | Gabbay               | N/A      | N/A         |
| 7806919 | 12/2009 | Bloom et al.         | N/A      | N/A         |
| 7815673 | 12/2009 | Bloom et al.         | N/A      | N/A         |
| 7824443 | 12/2009 | Salahieh et al.      | N/A      | N/A         |
| 7837727 | 12/2009 | Goetz                | 623/2.18 | A61F 2/2439 |
| 7892281 | 12/2010 | Seguin et al.        | N/A      | N/A         |
| 7914569 | 12/2010 | Nguyen et al.        | N/A      | N/A         |
| 7914575 | 12/2010 | Guyenot et al.       | N/A      | N/A         |
| 7959672 | 12/2010 | Salahieh et al.      | N/A      | N/A         |
| 7972378 | 12/2010 | Tabor et al.         | N/A      | N/A         |
| 7981123 | 12/2010 | Seguin               | N/A      | N/A         |
| 7981151 | 12/2010 | Rowe                 | N/A      | N/A         |
| 7988724 | 12/2010 | Salahieh et al.      | N/A      | N/A         |
| 7993392 | 12/2010 | Righini et al.       | N/A      | N/A         |
| 7993394 | 12/2010 | Hariton et al.       | N/A      | N/A         |
| 8007992 | 12/2010 | Tian et al.          | N/A      | N/A         |
| 8016877 | 12/2010 | Seguin et al.        | N/A      | N/A         |
| 8029556 | 12/2010 | Rowe                 | N/A      | N/A         |
| 8048153 | 12/2010 | Salahieh et al.      | N/A      | N/A         |
| 8052750 | 12/2010 | Tuval et al.         | N/A      | N/A         |



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| 8070800 | 12/2010 | Lock et al.        | N/A      | N/A         |
| 8070802 | 12/2010 | Lamphere et al.    | N/A      | N/A         |
| 8070805 | 12/2010 | Vidlund et al.     | N/A      | N/A         |
| 8075615 | 12/2010 | Eberhardt et al.   | N/A      | N/A         |
| 8080054 | 12/2010 | Rowe               | N/A      | N/A         |
| 8092520 | 12/2011 | Quadri             | N/A      | N/A         |
| 8092521 | 12/2011 | Figulla et al.     | N/A      | N/A         |
| 8096985 | 12/2011 | Legaspi et al.     | N/A      | N/A         |
| 8104149 | 12/2011 | McGarity           | N/A      | N/A         |
| 8109996 | 12/2011 | Stacchino et al.   | N/A      | N/A         |
| 8118866 | 12/2011 | Herrmann et al.    | N/A      | N/A         |
| 8133239 | 12/2011 | Oz et al.          | N/A      | N/A         |
| 8136218 | 12/2011 | Millwee et al.     | N/A      | N/A         |
| 8137398 | 12/2011 | Tuval et al.       | N/A      | N/A         |
| 8147542 | 12/2011 | Maisano et al.     | N/A      | N/A         |
| 8157852 | 12/2011 | Bloom et al.       | N/A      | N/A         |
| 8167932 | 12/2011 | Bourang et al.     | N/A      | N/A         |
| 8167934 | 12/2011 | Styrc              | 623/2.18 | A61F 2/2418 |
| 8172856 | 12/2011 | Eigler et al.      | N/A      | N/A         |
| 8182528 | 12/2011 | Salahieh et al.    | N/A      | N/A         |
| 8182530 | 12/2011 | Huber              | N/A      | N/A         |
| 8197528 | 12/2011 | Colgan et al.      | N/A      | N/A         |
| 8206437 | 12/2011 | Bonhoeffer et al.  | N/A      | N/A         |
| 8216174 | 12/2011 | Wilk et al.        | N/A      | N/A         |
| 8216301 | 12/2011 | Bonhoeffer et al.  | N/A      | N/A         |
| 8219229 | 12/2011 | Cao et al.         | N/A      | N/A         |
| 8220121 | 12/2011 | Hendriksen et al.  | N/A      | N/A         |
| 8221493 | 12/2011 | Boyle et al.       | N/A      | N/A         |
| 8226710 | 12/2011 | Nguyen et al.      | N/A      | N/A         |
| 8236045 | 12/2011 | Benichou et al.    | N/A      | N/A         |
| 8246675 | 12/2011 | Zegdi              | N/A      | N/A         |
| 8246678 | 12/2011 | Salahieh et al.    | N/A      | N/A         |
| 8252051 | 12/2011 | Chau et al.        | N/A      | N/A         |
| 8252052 | 12/2011 | Salahieh et al.    | N/A      | N/A         |
| 8287584 | 12/2011 | Salahieh et al.    | N/A      | N/A         |
| 8303653 | 12/2011 | Bonhoeffer et al.  | N/A      | N/A         |
| 8313525 | 12/2011 | Tuval et al.       | N/A      | N/A         |
| 8317858 | 12/2011 | Straubinger et al. | N/A      | N/A         |
| 8323335 | 12/2011 | Rowe et al.        | N/A      | N/A         |
| 8337541 | 12/2011 | Quadri et al.      | N/A      | N/A         |
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| 8348996 | 12/2012 | Tuval et al.       | N/A      | N/A         |
| 8353953 | 12/2012 | Giannetti et al.   | N/A      | N/A         |
| 8398704 | 12/2012 | Straubinger et al. | N/A      | N/A         |
| 8403983 | 12/2012 | Quadri et al.      | N/A      | N/A         |
| 8414643 | 12/2012 | Tuval et al.       | N/A      | N/A         |
| 8414644 | 12/2012 | Quadri et al.      | N/A      | N/A         |
| 8414645 | 12/2012 | Dwork et al.       | N/A      | N/A         |
| 8416643 | 12/2012 | Magee              | N/A      | N/A         |
| 8425404 | 12/2012 | Wilson et al.      | N/A      | N/A         |
| 8444689 | 12/2012 | Zhang              | N/A      | N/A         |
| 8449599 | 12/2012 | Chau et al.        | N/A      | N/A         |

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| 8449606 | 12/2012 | Eliassen et al.     | N/A      | N/A         |
| 8449625 | 12/2012 | Campbell et al.     | N/A      | N/A         |
| 8454685 | 12/2012 | Hariton et al.      | N/A      | N/A         |
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| 8460368 | 12/2012 | Taylor et al.       | N/A      | N/A         |
| 8460370 | 12/2012 | Zakay               | N/A      | N/A         |
| 8470023 | 12/2012 | Eidenschink et al.  | N/A      | N/A         |
| 8470028 | 12/2012 | Thornton et al.     | N/A      | N/A         |
| 8475521 | 12/2012 | Suri et al.         | N/A      | N/A         |
| 8475523 | 12/2012 | Duffy               | N/A      | N/A         |
| 8479380 | 12/2012 | Malewicz et al.     | N/A      | N/A         |
| 8480730 | 12/2012 | Maurer et al.       | N/A      | N/A         |
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| 8511244 | 12/2012 | Holecek et al.      | N/A      | N/A         |
| 8512401 | 12/2012 | Murray, III et al.  | N/A      | N/A         |
| 8518096 | 12/2012 | Nelson              | N/A      | N/A         |
| 8518106 | 12/2012 | Duffy et al.        | N/A      | N/A         |
| 8535368 | 12/2012 | Headley, Jr. et al. | N/A      | N/A         |
| 8540767 | 12/2012 | Zhang               | N/A      | N/A         |
| 8562663 | 12/2012 | Mearns et al.       | N/A      | N/A         |
| 8579963 | 12/2012 | Tabor               | N/A      | N/A         |
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| 8584849 | 12/2012 | McCaffrey           | N/A      | N/A         |
| 8585749 | 12/2012 | Shelso              | N/A      | N/A         |
| 8585755 | 12/2012 | Chau et al.         | N/A      | N/A         |
| 8585756 | 12/2012 | Bonhoeffer et al.   | N/A      | N/A         |
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| 8652202 | 12/2013 | Alon et al.         | N/A      | N/A         |
| 8652203 | 12/2013 | Quadri              | 623/2.11 | A61F 2/2418 |
| 8668733 | 12/2013 | Haug et al.         | N/A      | N/A         |
| 8673000 | 12/2013 | Tabor et al.        | N/A      | N/A         |
| 8679174 | 12/2013 | Ottma et al.        | N/A      | N/A         |
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| 8721665 | 12/2013 | Oz et al.           | N/A      | N/A         |
| 8721708 | 12/2013 | Seguin et al.       | N/A      | N/A         |
| 8721714 | 12/2013 | Kelley              | N/A      | N/A         |
| 8728154 | 12/2013 | Alkhatib            | N/A      | N/A         |
| 8728155 | 12/2013 | Montorfano et al.   | N/A      | N/A         |
| 8740918 | 12/2013 | Seguin              | N/A      | N/A         |
| 8740974 | 12/2013 | Lambrecht et al.    | N/A      | N/A         |
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| 8747458 | 12/2013 | Tuval et al.        | N/A      | N/A         |

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| 8747459 | 12/2013 | Nguyen et al.     | N/A      | N/A         |
| 8747460 | 12/2013 | Tuval et al.      | N/A      | N/A         |
| 8753384 | 12/2013 | Leanna            | N/A      | N/A         |
| 8758432 | 12/2013 | Solem             | N/A      | N/A         |
| 8764818 | 12/2013 | Gregg             | N/A      | N/A         |
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| 8771346 | 12/2013 | Tuval et al.      | N/A      | N/A         |
| 8771347 | 12/2013 | DeBoer et al.     | N/A      | N/A         |
| 8778017 | 12/2013 | Eliassen et al.   | N/A      | N/A         |
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| 8784337 | 12/2013 | Voeller et al.    | N/A      | N/A         |
| 8784478 | 12/2013 | Tuval et al.      | N/A      | N/A         |
| 8784481 | 12/2013 | Alkhatib et al.   | N/A      | N/A         |
| 8790387 | 12/2013 | Nguyen et al.     | N/A      | N/A         |
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| 8808356 | 12/2013 | Braido et al.     | N/A      | N/A         |
| 8828078 | 12/2013 | Salahieh et al.   | N/A      | N/A         |
| 8828079 | 12/2013 | Thielen et al.    | N/A      | N/A         |
| 8834564 | 12/2013 | Tuval et al.      | N/A      | N/A         |
| 8840663 | 12/2013 | Salahieh et al.   | N/A      | N/A         |
| 8845718 | 12/2013 | Tuval et al.      | N/A      | N/A         |
| 8845722 | 12/2013 | Gabbay            | 623/2.14 | A61F 2/2409 |
| 8858620 | 12/2013 | Salahieh et al.   | N/A      | N/A         |
| 8870948 | 12/2013 | Erzberger et al.  | N/A      | N/A         |
| 8870950 | 12/2013 | Hacohen           | N/A      | N/A         |
| 8876893 | 12/2013 | Dwork et al.      | N/A      | N/A         |
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| 8876895 | 12/2013 | Tuval et al.      | N/A      | N/A         |
| 8911455 | 12/2013 | Quadri et al.     | N/A      | N/A         |
| 8926693 | 12/2014 | Duffy et al.      | N/A      | N/A         |
| 8926694 | 12/2014 | Costello          | N/A      | N/A         |
| 8939960 | 12/2014 | Rosenman et al.   | N/A      | N/A         |
| 8945177 | 12/2014 | Dell et al.       | N/A      | N/A         |
| 8945209 | 12/2014 | Bonyuet et al.    | N/A      | N/A         |
| 8951299 | 12/2014 | Paul et al.       | N/A      | N/A         |
| 8961583 | 12/2014 | Hojeibane et al.  | N/A      | N/A         |
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| 8961595 | 12/2014 | Alkhatib          | N/A      | N/A         |
| 8968395 | 12/2014 | Hauser et al.     | N/A      | N/A         |
| 8974524 | 12/2014 | Yeung et al.      | N/A      | N/A         |
| 8979922 | 12/2014 | Jayasinghe et al. | N/A      | N/A         |
| 8986372 | 12/2014 | Murry, III et al. | N/A      | N/A         |
| 8986375 | 12/2014 | Garde et al.      | N/A      | N/A         |
| 8992608 | 12/2014 | Haug et al.       | N/A      | N/A         |
| 8998979 | 12/2014 | Seguin et al.     | N/A      | N/A         |
| 8998980 | 12/2014 | Shipley et al.    | N/A      | N/A         |
| 9005270 | 12/2014 | Perkins et al.    | N/A      | N/A         |
| 9005273 | 12/2014 | Salahieh et al.   | N/A      | N/A         |
| 9011521 | 12/2014 | Haug et al.       | N/A      | N/A         |
| 9011523 | 12/2014 | Seguin            | N/A      | N/A         |
| 9011524 | 12/2014 | Eberhardt         | N/A      | N/A         |

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| 9028545 | 12/2014 | Taylor             | N/A | N/A         |
| 9029418 | 12/2014 | Dove et al.        | N/A | N/A         |
| 9034032 | 12/2014 | McLean et al.      | N/A | N/A         |
| 9055937 | 12/2014 | Rowe et al.        | N/A | N/A         |
| 9066801 | 12/2014 | Kovalsky et al.    | N/A | N/A         |
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| 9078751 | 12/2014 | Naor               | N/A | N/A         |
| 9084676 | 12/2014 | Chau et al.        | N/A | N/A         |
| 9125738 | 12/2014 | Figulla et al.     | N/A | N/A         |
| 9138312 | 12/2014 | Tuval et al.       | N/A | N/A         |
| 9161834 | 12/2014 | Taylor et al.      | N/A | N/A         |
| 9173737 | 12/2014 | Hill et al.        | N/A | N/A         |
| 9180004 | 12/2014 | Alkhatib           | N/A | N/A         |
| 9186249 | 12/2014 | Rolando et al.     | N/A | N/A         |
| 9198757 | 12/2014 | Schroeder et al.   | N/A | N/A         |
| 9220507 | 12/2014 | Patel et al.       | N/A | N/A         |
| 9220594 | 12/2014 | Braido et al.      | N/A | N/A         |
| 9241790 | 12/2015 | Lane et al.        | N/A | N/A         |
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| 9259317 | 12/2015 | Wilson et al.      | N/A | N/A         |
| 9277990 | 12/2015 | Klima et al.       | N/A | N/A         |
| 9277993 | 12/2015 | Gamarra et al.     | N/A | N/A         |
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| 9289296 | 12/2015 | Braido et al.      | N/A | N/A         |
| 9295551 | 12/2015 | Straubinger et al. | N/A | N/A         |
| 9295552 | 12/2015 | McLean             | N/A | A61F 2/2409 |
| 9301834 | 12/2015 | Tuval et al.       | N/A | N/A         |
| 9308360 | 12/2015 | Bishop et al.      | N/A | N/A         |
| 9331328 | 12/2015 | Eberhardt et al.   | N/A | N/A         |
| 9339382 | 12/2015 | Tabor et al.       | N/A | N/A         |
| 9351831 | 12/2015 | Braido et al.      | N/A | N/A         |
| 9351832 | 12/2015 | Braido et al.      | N/A | N/A         |
| 9364321 | 12/2015 | Alkhatib et al.    | N/A | N/A         |
| 9387071 | 12/2015 | Tuval et al.       | N/A | N/A         |
| 9427327 | 12/2015 | Parrish            | N/A | N/A         |
| 9439763 | 12/2015 | Geist et al.       | N/A | N/A         |
| 9445897 | 12/2015 | Bishop et al.      | N/A | N/A         |
| 9456877 | 12/2015 | Weitzner et al.    | N/A | N/A         |
| 9510837 | 12/2015 | Seguin             | N/A | N/A         |
| 9510946 | 12/2015 | Chau et al.        | N/A | N/A         |
| 9572660 | 12/2016 | Braido et al.      | N/A | N/A         |
| 9597181 | 12/2016 | Christianson       | N/A | A61F 2/2448 |
| 9610159 | 12/2016 | Christianson       | N/A | A61F 2/2415 |
| 9642704 | 12/2016 | Tuval et al.       | N/A | N/A         |
| 9681951 | 12/2016 | Ratz               | N/A | A61F 2/2418 |
| 9681968 | 12/2016 | Goetz et al.       | N/A | N/A         |
| 9687345 | 12/2016 | Rabito et al.      | N/A | N/A         |
| 9700329 | 12/2016 | Metzger et al.     | N/A | N/A         |
| 9700411 | 12/2016 | Klima et al.       | N/A | N/A         |
| 9700445 | 12/2016 | Martin et al.      | N/A | N/A         |
| 9724083 | 12/2016 | Quadri et al.      | N/A | N/A         |

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| 9730790  | 12/2016 | Quadri et al.            | N/A | N/A          |
| 9730791  | 12/2016 | Ratz                     | N/A | A61F 2/2418  |
| 9775963  | 12/2016 | Miller                   | N/A | N/A          |
| 9795479  | 12/2016 | Lim et al.               | N/A | N/A          |
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*Primary Examiner: Mathew; Seema*

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## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/199,884, filed May 19, 2023, which is a continuation of U.S. patent application Ser. No. 18/109,075, filed Feb. 13, 2023, which is a continuation of U.S. patent application Ser. No. 17/008,231, filed Aug. 31, 2020, now U.S. Pat. No. 11,583,396, which is a continuation of U.S. patent application Ser. No. 16/171,263, filed Oct. 25, 2018, now U.S. Pat. No. 10,758,342, which is a continuation of U.S. patent application Ser. No. 15/455,713, filed Mar. 10, 2017, now U.S. Pat. No. 10,111,748, which is a continuation of U.S. patent application Ser. No. 15/227,238, filed Aug. 3, 2016, now U.S. Pat. No. 9,717,591, which is a continuation of U.S. patent application Ser. No. 14/801,713, filed Jul. 16, 2015, now U.S. Pat. No. 9,433,500, which is a continuation of U.S. patent application Ser. No. 14/255,179, filed Apr. 17, 2014, now U.S. Pat. No. 9,084,676, which is a continuation of U.S. patent application Ser. No. 14/025,594, filed Sep. 12, 2013, now U.S. Pat. No. 8,926,691, which is a continuation of U.S. patent application Ser. No. 13/597,122, filed Aug. 28, 2012, now U.S. Pat. No. 8,585,755, which is a continuation of U.S. patent application Ser. No. 12/959,292, filed Dec. 2, 2010, now U.S. Pat. No. 8,449,599, which claims the benefit of U.S. Provisional Application Nos. 61/287,099, filed Dec. 16, 2009, and 61/266,774, filed Dec. 4, 2009. Each related application is incorporated by reference herein in their entireties.

## **FIELD**

(1) This disclosure pertains generally to prosthetic devices for repairing and/or replacing native heart valves, and in particular to prosthetic valves for replacing defective mitral valves, as well as methods and devices for delivering and implanting the same within a human heart.

## **BACKGROUND**

(2) Prosthetic valves have been used for many years to treat cardiac valvular disorders. The native heart

valves (i.e., the aortic, pulmonary, tricuspid and mitral valves) serve critical functions in assuring the forward flow of an adequate supply of blood through the cardiovascular system. These heart valves can be rendered less effective by congenital malformations, inflammatory processes, infectious conditions or disease. Such damage to the valves can result in serious cardiovascular compromise or death. For many years the definitive treatment for such disorders was the surgical repair or replacement of the valve during open heart surgery. However, such surgeries are highly invasive and are prone to many complications. Therefore, elderly and frail patients with defective heart valves often go untreated. More recently a transvascular technique has been developed for introducing and implanting a prosthetic heart valve using a flexible catheter in a manner that is much less invasive than open heart surgery.

(3) In this technique, a prosthetic valve is mounted in a crimped state on the end portion of a flexible catheter and advanced through a blood vessel of the patient until the valve reaches the implantation site. The valve at the catheter tip is then expanded to its functional size at the site of the defective native valve such as by inflating a balloon on which the valve is mounted.

(4) Another known technique for implanting a prosthetic aortic valve is a transapical approach where a small incision is made in the chest wall of a patient and the catheter is advanced through the apex (i.e., bottom tip) of the heart. Transapical techniques are disclosed in U.S. Patent Application Publication No. 2007/0112422, which is hereby incorporated by reference. Like the transvascular approach, the transapical approach can include a balloon catheter having a steering mechanism for delivering a balloon-expandable prosthetic heart valve through an introducer to the aortic annulus. The balloon catheter can include a deflecting segment just proximal to the distal balloon to facilitate positioning of the prosthetic heart valve in the proper orientation within the aortic annulus.

(5) The above techniques and others have provided numerous options for high operative risk patients with aortic valve disease to avoid the consequences of open heart surgery and cardiopulmonary bypass. While devices and procedures for the aortic valve are well-developed, such catheter-based procedures are not necessarily applicable to the mitral valve due to the distinct differences between the aortic and mitral valve. The mitral valve has complex subvalvular apparatus, i.e., chordae tendinae, which are not present in the aortic valve.

(6) Surgical mitral valve repair techniques (e.g., mitral annuloplasty) have increased in popularity due to their high success rates, and clinical improvements noted after repair. In addition to the existing mitral valve repair technologies, there are a number of new technologies aimed at making mitral valve repair a less invasive procedure. These technologies range from iterations of the Alfieri stitch procedure to coronary sinus-based modifications of mitral anatomy to subvalvular plications or ventricular remodeling devices, which would incidentally correct mitral regurgitation.

(7) However, for mitral valve replacement, few less-invasive options are available. There are approximately 25,000 mitral valve replacements (MVR) each year in the United States. However, it is estimated that over 300,000 patients who meet guidelines for treatment are denied treatment based on their age and/or co-morbidities. Thus, a need exists for minimally invasive techniques for replacing the mitral valve.

## SUMMARY

(8) Prosthetic mitral valves, components thereof, and methods and devices for implanting the same are described herein.

(9) A prosthetic apparatus is described that is configured for implanting at the native mitral valve region of the heart and includes a main body that is radially compressible to a radially compressed state and self-expandable from the compressed state to a radially expanded state. The prosthetic apparatus also comprises at least one ventricular anchor coupled to the main body and disposed outside of the main body such that when the main body is compressed to the compressed state, a leaflet-receiving space between the ventricular anchor and an outer surface of the main body increases to receive a native valve leaflet therebetween. When the main body self-expands to the expanded state in the absence of any substantial external inward forces on the main body or the ventricular anchor, the space decreases to capture the leaflet between the main body and the ventricular anchor.

(10) In some embodiments, a prosthetic apparatus, for implanting at the native mitral valve region of the heart, includes a frame having a main body and at least one ventricular anchor coupled to and disposed

outside of the main body. The prosthetic apparatus also includes a plurality of leaflets supported by the main body that form a one-way valve for the flow of blood through the main body. The main body is radially compressible to a radially compressed state for delivery into the body and self-expandable from the compressed state to a radially expanded state. The ventricular anchor comprises a base that is fixedly secured to the main body, a free end portion opposite the base, and an intermediate portion defining a leaflet-receiving space between the ventricular anchor and the main body for receiving a leaflet of the native valve. Expansion of the main body from its compressed state to its radially expanded state in the absence of any radial inward forces on the ventricular anchor causes the leaflet-receiving space to decrease.

(11) In other embodiments, a prosthetic apparatus for implanting at the native mitral valve region includes a main body, at least one ventricular anchor and at least one atrial anchor. The main body is configured for placement within the native mitral valve and is compressible to a compressed state for delivery into the heart and self-expandable from the compressed state to an expanded state. At least one ventricular anchor is coupled to and disposed outside of the main body such that, in the expanded state, a leaflet-receiving space exists between the ventricular anchor and an outer surface of the main body to receive a free edge portion of a native valve leaflet. The ventricular anchor comprises an engagement portion configured to extend behind the received native leaflet and contact a ventricular surface of the native mitral annulus, the annulus connection portion of the received native leaflet, or both the ventricular surface of the native annulus and the annulus connection portion of the received native leaflet. At least one atrial sealing member is coupled to and disposed outside of the main body and is configured to contact an atrial portion of the native mitral annulus, the annulus connection portion of the received native leaflet, or both the atrial surface of the native annulus and the annulus connection portion of the received native leaflet at a location opposite from the engagement portion of the ventricular anchor for retention of the prosthetic apparatus and/or prevention of paravalvular leakage.

(12) Exemplary delivery systems are also described for delivering a prosthetic apparatus into the heart. Some embodiments include an inner sheath having a distal end portion having at least one longitudinal slot extending proximally from a distal end of the inner sheath. The distal end portion of the inner sheath is configured to contain the prosthetic apparatus in a radially compressed state. An outer sheath is positioned concentrically around the inner sheath and at least one of the inner sheath and outer sheath is movable axially relative to the other between a first position in which the outer sheath extends over at least a portion of the longitudinal slot and a second position in which the at least a portion of the longitudinal slot is uncovered by the outer sheath so to allow a portion of the prosthetic apparatus contained within the inner sheath to expand radially outward through the slot.

(13) Exemplary methods are also described for implanting a prosthetic apparatus at the native mitral valve region of the heart. One such method includes delivering the prosthetic apparatus into the heart in a radially compressed state; allowing a ventricular anchor to self-expand away from a main body of the frame while the main body is held in the compressed state, thereby increasing a gap between the ventricular anchor and an outer surface of the main body; positioning the main body in the annulus of the native mitral valve and the ventricular anchor adjacent the ventricular side of a native mitral valve leaflet such that the leaflet is disposed in the gap between the ventricular anchor and the outer surface of the main body; and allowing the main body to self-expand to an expanded state such that the gap decreases to capture the leaflet between the outer surface of the main body and the ventricular anchor.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross sectional view of the human heart.
- (2) FIG. 2 is another cross sectional view of the human heart showing the mitral valve region.
- (3) FIG. 3 is a schematic view of the native mitral valve anatomy showing the mitral leaflets attached to the papillary muscles via chordae tendineae.
- (4) FIG. 4A is a diagram of native mitral valve showing Carpentier nomenclature.
- (5) FIG. 4B shows a native mitral valve with a gap between the leaflets.

- (6) FIGS. 4C and 4D show an exemplary prosthetic valve positioned within a native mitral valve.
- (7) FIG. 5 is a side view of an exemplary embodiment of a prosthetic valve.
- (8) FIG. 6 shows the prosthetic valve of FIG. 5 rotated 90 degrees with respect to a longitudinal axis of the valve.
- (9) FIG. 7 is a ventricular (outflow) view of the prosthetic valve shown of FIG. 5.
- (10) FIGS. 8-10 are views corresponding to FIGS. 5-7, showing an exemplary embodiment of a frame of the prosthetic valve of FIGS. 5-7.
- (11) FIGS. 11-16 are a series of side views of the frame of FIGS. 9, without the atrial sealing member, showing the leaflet-receiving spaces between the ventricular anchors and the main body increasing as the main body is radially compressed.
- (12) FIGS. 17-22 are a series of end views corresponding to FIGS. 11-16, respectively.
- (13) FIG. 23 is a cross-sectional view of the heart showing the frame of FIG. 9 implanted in the mitral valve region, wherein the native mitral valve leaflets are captured between the main body and the ventricular anchors.
- (14) FIG. 24 shows exemplary dimensions of the atrial sealing member, main body and ventricular anchors of FIG. 9.
- (15) FIG. 25 shows an exemplary embodiment of a frame, with the atrial sealing member excluded, comprising a “T” shaped pushing member extending downward from a ventricular end of the main body.
- (16) FIG. 26 shows an exemplary embodiment of a frame, with the atrial sealing member excluded, comprising a “V” shaped pushing member extending downward from the ventricular end of the main body.
- (17) FIGS. 27-29 show an exemplary embodiment of a prosthetic valve having a frame with four ventricular anchors.
- (18) FIGS. 30-32 show the frame of the prosthetic valve shown in FIGS. 27-29.
- (19) FIG. 33 is a cross-sectional view of the heart showing the frame of FIGS. 30-32 implanted in the mitral valve region.
- (20) FIG. 34 is a cross-sectional view of the heart showing an embodiment of a frame, comprising extended ventricular anchors and an atrial sealing member, implanted in the mitral valve region such that the mitral annulus and/or native leaflets are compressed between the ends of the extended ventricular anchors and the atrial sealing member.
- (21) FIGS. 35 and 36 are side views of an exemplary embodiment of a frame comprising “S” shaped ventricular anchors.
- (22) FIGS. 37 and 38 are side and top views, respectively, of an exemplary embodiment of a frame, with the atrial sealing member excluded, comprising wider shaped ventricular anchors.
- (23) FIG. 39 is a cross-sectional view of the heart showing an embodiment of a frame implanted in the mitral valve region, wherein the ventricular anchors remain separated from the body of the frame after expansion and the ventricular anchors contact the lower ends of the mitral leaflets to utilize tension from the chordae tendineae to retain the frame.
- (24) FIG. 40 shows an exemplary embodiment of a frame comprising a substantially flat atrial sealing member.
- (25) FIG. 41 shows an exemplary embodiment of a frame comprising an upwardly extending atrial sealing member.
- (26) FIG. 42 shows an exemplary embodiment of a frame comprising an upwardly extending atrial sealing member and extended ventricular anchors.
- (27) FIG. 43 shows an exemplary embodiment of a frame, with the atrial sealing member excluded, comprising wide-set ventricular anchors.
- (28) FIG. 44 depicts a series of side views of an exemplary embodiment of a frame, with the atrial sealing member excluded, having ventricular anchors that flip up into a final configuration.
- (29) FIG. 45 depicts a series of side views of an exemplary embodiment of a frame, with the atrial sealing member excluded, having ventricular anchors that curl up into a final configuration.
- (30) FIGS. 46A-46C show an exemplary embodiment of a frame, with the atrial sealing member

excluded, wherein the main body is manufactured separately from the ventricular anchors.

(31) FIGS. **47A-47D** show another embodiment of a frame, with the atrial sealing member excluded, wherein the main body is manufactured separately from the ventricular anchors and attached using a sleeve.

(32) FIGS. **48A-48C** show another embodiment of a frame, with the atrial sealing member excluded, wherein the main body is manufactured separately from the ventricular anchors and attached using a sleeve with a mechanical lock.

(33) FIG. **49** shows an exemplary embodiment of a delivery system for delivering and implanting a prosthetic valve at a native mitral valve region of the heart.

(34) FIG. **50** is a detailed view of the distal portion of the delivery system of FIG. **49**.

(35) FIG. **51** is a cross-sectional view of a handle portion of the delivery system of FIG. **49**, taken along section line **51-51**.

(36) FIG. **52** is a cross sectional view of the handle portion of the delivery system of FIG. **49**, taken along section line **52-52**.

(37) FIG. **53** is a cross sectional view of an insertable portion of the delivery system of FIG. **49**, taken along section line **53-53**.

(38) FIG. **54** shows the delivery system of FIG. **49** with a prosthetic valve loaded within a slotted inner sheath with the ventricular anchors extending outward through slots of the inner sheath.

(39) FIG. **55** is a cross-sectional view of the delivery system of FIG. **49** in a delivery position containing the prosthetic valve within inner and outer sheaths and between a nose cone and a tip of a pusher shaft.

(40) FIG. **56** is a cross-sectional view of a distal end portion of the delivery system of FIG. **49** showing the outer sheath of the delivery system retracted such that ventricular anchors extend outward through slots of the inner sheath.

(41) FIG. **57** is a cross-sectional view of the heart showing the ventricular anchors of the prosthetic valve being pre-deployed in the left ventricle using the delivery system of FIG. **49**.

(42) FIG. **58** is a view of the mitral valve region of the heart from the left ventricle showing the ventricular anchors extending from the slots in the delivery system and showing the ventricular anchors positioned between respective mitral leaflets and the ventricular walls.

(43) FIG. **59** is a cross-sectional view of the heart showing the prosthetic valve being implanted in the mitral valve region using the delivery system of FIG. **49** with the native leaflets positioned between the ventricular anchors and the inner sheath.

(44) FIG. **60** is a cross-sectional view of the delivery system of FIG. **49** showing the slotted inner sheath retracted to a point where the ventricular anchors of the prosthetic valve contact a notched retaining band around the slotted inner sheath.

(45) FIG. **61** is a cross-sectional view of the delivery system of FIG. **49** showing the slotted inner sheath fully retracted after the band has been broken, and the prosthetic valve in an expanded state after being fully deployed from the sheath.

(46) FIG. **62** is a view of the mitral valve region of the heart from the left ventricle showing an exemplary embodiment of a prosthetic valve fully implanted with the mitral leaflets captured between a main body and ventricular anchors.

(47) FIG. **63** shows an exemplary embodiment of a prosthetic valve within a catheter sheath for delivering to a native valve region of the heart, according to another embodiment.

(48) FIG. **64** shows the prosthetic valve of FIG. **63** with the catheter sheath pulled back such that the ventricular anchors are free to expand but the main body remains compressed.

(49) FIG. **65** shows the prosthetic valve of FIG. **63** with the outer sheath recapturing the main body such that only the ventricular anchors are exposed.

(50) FIG. **66** is a cross-sectional view of the heart showing the prosthetic valve of FIG. **65** being implanted in the native mitral valve region using a transatrial approach.

(51) FIG. **67** is a cross-sectional view of the heart showing the prosthetic valve of FIG. **65** being implanted in the native mitral valve region using a transeptal approach.

(52) FIG. **68** is a view of the mitral valve region from the left ventricle showing an embodiment of an atrially delivered prosthetic valve having ventricular anchors extending free of a sheath and positioned



between the native mitral valve leaflets and the ventricular walls.

(53) FIG. **69** is a view of the mitral valve region from the left ventricle showing the prosthetic valve of FIG. **68** fully expanded and anchored to the native mitral valve leaflets.

(54) FIG. **70** is a cross-sectional view of the heart showing an embodiment of a docking frame that is secured to the native tissue of mitral valve region and a separately deployed prosthetic valve that is secured to the docking frame within the lumen of the docking frame.

(55) FIG. **71** a perspective view of an embodiment of a prosthetic apparatus for implanting at the native mitral valve region to treat mitral regurgitation.

(56) FIG. **72** is a side view of the prosthetic apparatus of FIG. **71**.

(57) FIG. **73** is another side view of the prosthetic apparatus of FIG. **71**.

(58) FIG. **74** is an end view of the prosthetic apparatus of FIG. **71**.

(59) FIGS. **75-79** are cross-sectional views of the heart showing a transeptal delivery of the prosthetic apparatus of FIG. **71**.

(60) FIG. **80** is a side view of an alternative embodiment of a prosthetic apparatus of FIG. **71**, comprising prosthetic valve.

(61) FIG. **81** is a partial side view of an alternative embodiment of a prosthetic apparatus of FIG. **71**, comprising a Z-cut frame body.

(62) FIG. **82** is a partial side view of an alternative embodiment of a prosthetic apparatus of FIG. **71**, comprising a lattice frame body and a prosthetic valve.

(63) FIG. **83** is a partial side view of an alternative embodiment of a prosthetic apparatus of FIG. **71** comprising a helical frame body.

(64) FIGS. **84** and **85** show an exemplary method for implanting an exemplary prosthetic apparatus having “L” shaped ventricular anchors.

(65) FIGS. **86** and **87** show another exemplary method for implanting another prosthetic apparatus having “L” shaped ventricular anchors.

(66) FIG. **88** is ventricular view of the native mitral valve region.

#### DETAILED DESCRIPTION

(67) Described herein are embodiments of prosthetic valves and components thereof that are primarily intended to be implanted at the mitral valve region of a human heart, as well as apparatus and methods for implanting the same. The prosthetic valves can be used to help restore and/or replace the functionality of a defective native valve.

(68) The Human Heart

(69) Relevant portions of the human heart are shown in FIGS. **1** and **2**. A healthy heart has a generally conical shape that tapers to a lower apex **38**. The heart is four-chambered and comprises the left atrium **4**, right atrium **26**, left ventricle **6**, and right ventricle **28**. The left and right sides of the heart are separated by a wall generally referred to as the septum **30**. The native mitral valve **2** of the human heart connects the left atrium **4** to the left ventricle **6**. The mitral valve **2** has a very different anatomy than other native heart valves, such as the aortic valve **14**.

(70) The mitral valve **2** includes an annulus portion **8**, which is an annular portion of the native valve tissue surrounding the mitral valve orifice, and a pair of cusps, or leaflets, **10**, **12** extending downward from the annulus **8** into the left ventricle **6**. The mitral valve annulus **8** can form a “D” shaped, oval, or otherwise out-of-round cross-sectional shape having major and minor axes. The anterior leaflet **10** can be larger than the posterior leaflet **12**, as shown schematically in FIG. **4A**, forming a generally “C” shaped boundary between the abutting free edges of the leaflets when they are closed together. FIG. **4B** shows the native mitral valve **2** with a slight gap **3** between the leaflets **10**, **12**, such as with a defective native mitral valve that fails to completely close, which can lead to mitral regurgitation and/or other undesirable conditions.

(71) When operating properly, the anterior leaflet **10** and the posterior leaflet **12** function together as a one-way valve to allow blood to flow only from the left atrium **4** to the left ventricle **6**. The left atrium **4** receives oxygenated blood from the pulmonary veins **32**. When the muscles of the left atrium **4** contract and the left ventricle dilates, the oxygenated blood that is collected in the left atrium **4** flows into the left ventricle **6**. When the muscles of the left atrium **4** relax and the muscles of the left ventricle **6** contract,

the increased blood pressure in the left ventricle urges the two leaflets together, thereby closing the one-way mitral valve so that blood cannot flow back to the left atrium and is instead expelled out of the left ventricle through the aortic valve **14**.

(72) To prevent the two leaflets **10**, **12** from prolapsing under pressure and folding back through the mitral annulus **8** toward the left atrium **4**, a plurality of fibrous cords called chordae tendineae **16** tether the leaflets **10**, **12** to papillary muscles in the left ventricle **6**. Referring to FIGS. **3** and **4**, chordae **16** are attached to and extend between the postero-medial papillary muscle **22** and the postero-medial margins of both the anterior leaflet **10** and the posterior leaflet **12** (A1 and P1 areas, respectively, as identified by Carpentier nomenclature). Similarly, chordae **16** are attached to and extend between the antero-lateral papillary muscle **24** and the antero-lateral margins of both the anterior leaflet **10** and the posterior leaflet **12** (A3 and P3 areas, respectively, as identified by Carpentier nomenclature). The A2 and P2 areas are relatively free of chordae attachment points and provide a region where a prosthetic mitral valve can be anchored (see FIG. **3**). In addition, the organization of the chordae provides an approach path to deliver a prosthetic mitral valve with minimal risk of chordae entanglement.

#### (73) Prosthetic Valve

(74) When the native mitral valve fails to function properly, a prosthetic valve replacement can help restore the proper functionality. Compared to the aortic valve **14**, however, which has a relatively round and firm annulus (especially in the case of aortic stenosis), the mitral valve annulus **8** can be relatively less firm and more unstable. Consequently, it may not be possible to secure a prosthetic valve that is designed primarily for the aortic valve within the native mitral valve annulus **8** by relying solely on friction from the radial force of an outer surface of a prosthetic valve pressed against the native mitral annulus **8**. Accordingly, the prosthetic valves described herein can rely on ventricular anchors instead of, or in addition to, radial friction forces, to secure the prosthetic valve within the native mitral valve annulus **8** (see FIG. **23**, for example).

(75) In addition to providing an anchoring means for the prosthetic valve, the ventricular anchors can also remodel the left ventricle **6** to help treat an underlying cause of mitral regurgitation—left ventricle enlargement/dilation. The ventricular anchors can pull the native mitral valve leaflets **10**, **12** closer together and toward the left atrium and, via the chordae **16**, thereby pull the papillary muscles **22**, **24** closer together, which can positively remodel the ventricle acutely and prevent the left ventricle from further enlarging. Thus, the ventricular anchors can also be referred to as tensioning members or reshaping members.

(76) FIGS. **5-7** illustrate an exemplary prosthetic valve **100**, according to one embodiment, that can be implanted in the native mitral valve region of the heart to replace the functionality of the native mitral valve **2**. The prosthetic valve **100** comprises a frame **102** and a valve structure **104** supported by and/or within the frame. The valve structure **104** can include a plurality of prosthetic leaflets **106** (three in the illustrated embodiment) and/or other components for regulating the flow of blood in one direction through the prosthetic valve **100**. In FIGS. **5** and **6**, for example, valve structure **104** is oriented within the frame **102** such that an upper end **110** of the valve structure is the inflow end and a lower end **112** of the valve structure is the outflow end. The valve structure **104** can comprise any of various suitable materials, such as natural tissue (e.g., bovine pericardial tissue) or synthetic materials. The valve structure **104** can be mounted to the frame **102** using suitable techniques and mechanisms. In the illustrated embodiment, for example, the leaflets **106** are sutured to the frame **102** in a tricuspid arrangement, as shown in FIG. **7**.

(77) Additional details regarding components and assembly of prosthetic valves (including techniques for mounting leaflets to the frame) are described, for example, in U.S. Patent Application Publication No. 2009/0276040 A1 and U.S. patent application Ser. No. 12/393,010, which are incorporated by reference herein.

(78) As shown in FIGS. **8-10**, the frame **102** can comprise a tubular main body **122**, one or more ventricular anchors **126** extending from a ventricular end **130** of the main body and optionally an atrial sealing member **124** extending radially outward from an atrial end **132** of the main body. When the frame **102** is implanted in the native mitral valve region of the heart, as shown in FIG. **23**, the main body **122** is positioned within the native mitral valve annulus **8** with the ventricular end **130** of the main

body **122** being a lower outlet end, the atrial end **132** of the main body **132** being an upper inlet end, the ventricular anchors **126** being located in the left ventricle **6**, and the atrial sealing member **124** being located in the left atrium **4**.

(79) The frame **102** can be made of a wire mesh and can be radially collapsible and expandable between a radially expanded state and a radially compressed state (as shown schematically in a series of successive stages in FIGS. **11-16** and **17-22**) to enable delivery and implantation at the mitral valve region of the heart (or within another native heart valve). The embodiments of the frame **102** shown in FIGS. **11-22** do not include an atrial sealing member **124**, though other embodiments of the frame **102** do include an atrial sealing member **124**. The wire mesh can include metal wires or struts arranged in a lattice pattern, such as the sawtooth or zig-zag pattern shown in FIGS. **8-10** for example, but other patterns may also be used. The frame **102** can comprise a shape-memory material, such as Nitinol for example, to enable self-expansion from the radially compressed state to the expanded state. In alternative embodiments, the frame **102** can be plastically expandable from a radially compressed state to an expanded state by an expansion device, such as an inflatable balloon (not shown) for example. Such plastically expanding frames can comprise stainless steel, chromium alloys, and/or other suitable materials.

(80) In an expanded state, as shown in FIGS. **8-10**, the main body **122** of the frame **102** can form an open-ended tube. The valve structure **104** can be coupled to an inner surface of the frame **102**, such as via a material layer **142** on the inner surface of the frame, as discussed below, and can be retained within the lumen formed by the main body **122**, as shown in FIG. **7**. An outer surface of the main body **122** can have dimensions similar to that of the mitral orifice, i.e., the inner surface of the mitral annulus **8**, but not necessarily. In some embodiments, for example, the outer surface of the main body **122** can have diametrical dimensions that are smaller than the diametrical dimensions of the native mitral orifice, such that the main body **122** can fit within the mitral orifice in the expanded state without substantially stretching the native mitral annulus **8**, such as in FIG. **23**. In such embodiments, the frame **102** need not rely on a pressure fit, or friction fit, between the outer surface of the main body **122** and the inner surface of the mitral annulus **8** for prosthetic valve retention. Instead, the frame **102** can rely on the ventricular anchors **126** and/or the atrial sealing member **124** for retention, as further described below. In other embodiments, however, the main body **122** can be configured to expand to an equal or greater size than the native mitral orifice and thereby create a pressure fit when implanted.

(81) In embodiments wherein the main body **122** comprises diametrical dimensions that are smaller than the diametrical dimensions of the native mitral orifice, the main body can sit loosely, or “float,” between the native leaflets **10**, **12**. As shown in FIG. **4C**, this loose fit can create gaps **37** between the leaflets **10**, **12** and the main body **122** of the frame. To prevent blood flow between the outside of the prosthetic valve **100** and the native valve tissue, such as through the gaps **37**, the annular atrial sealing member **124** can create a fully annular contact area, or seal, with the native tissue on the atrial side of the mitral annulus **8**. Accordingly, as shown in FIG. **4D**, the atrial sealing member **124** can be sized to fully cover the gaps **37**.

(82) The ends of the frame **102** can have a sawtoothed or zig-zag pattern, as shown in FIGS. **8-10**, comprising a series of side-by-side “V” shaped portions connected together at their upper ends, for example. This pattern can facilitate compression and can help maximize a surface area with which the frame connects to the native tissue. Alternatively, the ends of the frame **102** can have a straight edge, or some other pattern.

(83) In some embodiments, the main body **122** can comprise at least one extension member, or pushing member, that extends downward from the ventricular end **130** of the main body **122**. The frame **202** shown in FIG. **25**, for example, comprises an extension member in the form of a prong **204** that extends from the lower vertex of one of the “V” shaped portions of a main body **222**. The prong **204** can have an upside-down “T” shape comprising a lower pushing surface **206**. In another embodiment, the frame **302** shown in FIG. **26** comprises a “V” shaped pushing member **304** that extends from two adjacent lower vertices of a main body **322** and comprises a pushing surface **306**. The pushing surfaces **206** and **306** can comprise the lowermost points on the frames **202** and **302**, respectively, and can provide a pushing surface for the frame to be expelled out of a delivery device without contacting the ventricular anchors

226, 326, as described in more detail below.

(84) With reference again to the embodiment shown in FIGS. 8-10, the atrial sealing member **124** of the frame **102** can be integral with the main body **122** and can be comprised of the same wire mesh lattice as the main body **122** such that the atrial sealing member **124** can also be radially collapsible and expandable. In the expanded state, the atrial sealing member **124** can be generally frustoconical and extend from the atrial end **132** of main body **122** both radially outward and axially downward toward the ventricular end **130** of the main body **122**. An outer rim **140** of the atrial sealing member **124** can be sized and shaped to contact the atrial side of the mitral annulus and tissue of the left atrium **8** when the frame **102** is implanted, as shown in FIG. 23. The end view profile of the outer rim **140**, as shown in FIG. 10, can have a generally circular, oval, or other shape that generally corresponds to the native geometry of the atrial walls **18** and the mitral annulus **8**. The contact between the atrial sealing member **124** and the tissue of the atrial walls **18** and/or the mitral annulus **8** can promote tissue ingrowth with the frame, which can improve retention and reduce paravalvular leakage.

(85) The atrial sealing member **124** desirably is sized such that when the prosthetic valve **100** is implanted in the native mitral valve, as shown in FIG. 23, the outer rim **140** contacts the native annulus **8** around the entire native valve and therefore completely covers the opening between the native leaflets **10, 12**. The atrial sealing member **124** desirably includes a sealing layer **142** that is impervious to the flow of blood. In this manner, the atrial sealing member **124** is able to block blood from flowing back into the left atrium between the outer surfaces of the prosthetic valve **100** and the native valve tissue. The atrial sealing member also ensures that all, or substantially all, of the blood passes through the one-way valve as it flows from the left atrium to the left ventricle.

(86) As shown in FIGS. 5-7, at least one biocompatible sheet or layer **142** can be connected to the inner and/or outer surfaces of the main body **122** and the atrial sealing member **124** to form at least one layer or envelope covering the openings in the wire mesh. The layer **142** can be connected to the frame **102** by sutures, for example. The layer **142** can form a fluid-occluding and/or sealing member that can at least partially block the flow of blood through the wire mesh to reduce paravalvular leakage and can promote tissue ingrowth with the frame **102**. The layer **142** can provide a mounting surface, or scaffold, to which the portions of the valve structure **104**, such as the leaflets **106**, can be secured. For example, the dashed line **108** in FIGS. 5 and 6 represents where the inlet ends of the leaflets **106** can be sewn, sutured, or otherwise secured to the layer **142**. This seam between the inlet ends of the leaflets **106** and the layer **142** can form a seal that is continuous around the inner perimeter of the layer **142** and can block blood flow between the inner surface of the layer **142** and the outer surface of the leaflets **106**. This seal can allow the prosthetic valve **100** to direct blood to flow between the plurality of leaflets **106**.

(87) The same layer **142** and/or one or more separate cuffs **144** can also wrap around, or cover, the end edges of the frame **102**, such as the ventricular end **130** of the main body **122** and/or the outer rim **140** of the atrial sealing member **124**. Such a cuff **144** can cover and protect sharp edges at the ends of the frame **102**. For example, in the embodiment shown in FIG. 5, the layer **142** extends from the outer rim **140** across the upper surface of the atrial sealing member **124** and downward along the inner surface of the main body **122** and comprises a cuff **144** that wraps around and covers a ventricular end portion of the main body **122**. The layer **142** can be sutured to the outer rim **140** and to the inner surface of the main body **122**.

(88) The layer **142** can comprise a semi-porous fabric that blocks blood flow but can allow for tissue ingrowth. The layer **142** can comprise synthetic materials, such as polyester material or a biocompatible polymer. One example of a polyester material is polyethylene terephthalate (PET). Alternative materials can be used. For example, the layer can comprise biological matter, such as natural tissue, pericardial tissue (e.g., bovine, porcine, or equine pericardium) or other biological tissue.

(89) With reference to FIGS. 8 and 9, one or more ventricular anchors **126** can extend from the main body **122** of the frame **102**, such as from the ventricular end **130** of the main body. The ventricular anchors **126** can function to retain the frame **102**, with or without the valve structure **104**, within a native valve region of the heart. In the embodiment shown in FIGS. 8 and 9, the frame **102** comprises two diametrically opposed ventricular anchors **126** that can function to secure the frame **102** to the anterior and posterior mitral leaflets **10, 12**, respectively, when the frame **102** is implanted in the mitral

valve region, as shown in FIG. 23. In alternate embodiments, the frame **102** can have three or more ventricular anchors **126**, which can be angularly spaced around the main body **122** of the frame.

(90) When the frame **102** is in an expanded state, as in FIG. 9, the geometry of the frame can cause the ventricular anchors **126** to be urged against the outer surface of the main body **122**. Alternatively, the ventricular anchors **126** can be configured to be spaced apart from the outer surface of the main body **122** when the frame **102** is in the expanded state (see FIG. 39, for example). In any case, when the frame **102** is radially compressed to the compressed state, the space or gap between the ventricular anchors **126** and the outer surface of the main body **122** can increase, as shown in FIGS. 11-16.

(91) While the main body **122** and the atrial sealing member **124** are in the compressed state, the frame **102** can be inserted into the mitral valve orifice such that the spaced apart ventricular anchors **126** wrap around the leaflets **10, 12** and extend upward between the leaflets and the ventricular walls **20** (see FIG. 59, for example). With reference to FIG. 23, an anterior ventricular anchor **146** can be located behind the anterior leaflet **10** and a posterior ventricular anchor **148** can be located behind the posterior leaflet **12**. With reference to FIGS. 3 and 4, the two ventricular anchors are desirably located behind the respective leaflets near the middle portions of the leaflets A2, P2 about midway between the commissures **36** where the two leaflets meet. These middle portions A2, P2 of the leaflets **10, 12** are desirable ventricular anchor locations because the chordae tendineae **16** attachments to the leaflets are sparser in these locations compared to locations nearer to the commissures **36**.

(92) When the main body **122** is subsequently expanded or allowed to self-expand to the expanded state, as shown in FIGS. 11-16 in reverse order, the ventricular anchors are configured to pivot radially inward relative to the main body **122**, without external compressive forces on the ventricular anchors. This causes the gaps between the ventricular anchors **126** and the outer surface of the main body **122** to decrease, thereby enabling the capture of the leaflets **10, 12** between the ventricular anchors and the main body. Conversely, compressing the main body **122** causes the ventricular anchors **126** to pivot away from the main body to increase the gaps between the outer surface of the main body and the ventricular anchors. In some embodiments, the free ends, or apexes, **162** of the ventricular anchors **126** can remain substantially the same distance apart from one another as the main body **122** is radially compressed or expanded free of external forces on the ventricular anchors. In some embodiments, such as the embodiment shown in FIG. 23, the frame is configured to compress the native mitral leaflets **10, 12** between the main body and the ventricular anchors when the frame expands to the expanded state. In other embodiments, such as the embodiment shown in FIG. 39, the ventricular anchors do not compress or clamp the native leaflets against the main body but still prevent the prosthetic valve from migrating toward the left atrium by the hooking of the ventricular anchors around the native leaflets **10, 12**. In such embodiments, the prosthetic valve **100** can be retained in place against migration toward the left ventricle by the atrial sealing member **124** as further described below.

(93) With reference to the embodiment shown in FIGS. 8-10, each ventricular anchor **126** can comprise a flexible, elongate member, or wire, **150** comprised of a shape memory material, such as, for example, Nitinol. In some embodiments, as shown in FIG. 8, each wire **150** can comprise a first end portion **152** coupled to a first attachment location **156** of the main body **122**, and a second end portion **154** coupled to a second attachment location **158** of the main body. The first and second end portions **152, 154** form a base of the ventricular anchor. The first and second attachment locations **152, 154** of the main body can be at, or adjacent to, the ventricular end **130** of the main body **122**. The two end portions **152, 154** of each wire **150** can be extensions of the wires or struts that make up the lattice mesh of the main body **122**. Each wire **150** further comprises an intermediate portion **160** extending in a direction lengthwise of the main body between the end portions **152, 154**. The intermediate portion **160** includes a bend **162** that forms the free end portion, or apex, of the ventricular anchor.

(94) The wire **150** can have a circular or non-circular cross-sectional profile perpendicular to a length of the wire, such as a polygonal cross-sectional profile. With reference to FIG. 8A, the wire **150** can comprise a rectangular cross-sectional shape having a length “L” and a relatively narrower width “W” such that when the two end portions **152, 154** of the ventricular anchor **126** attached to the frame **102** are moved toward each other, such as when the frame is compressed, the wire **150** bends primarily in the width direction. This promotes bending of the ventricular anchor **126** in a direction radially outward

away from the main body **122**, widening the gap between the ventricular anchor **126** and the main body **122**. This feature can help to capture a leaflet between the ventricular anchor **126** and the main body **122** during implantation.

(95) Ventricular anchors can comprise various shapes or configurations. Some frame embodiments, such as the frame **102** shown in FIG. **8**, comprise generally “U” or “V” shaped ventricular anchors **126** that connect to the main body **122** at two attachment locations **156**, **158**. The upper apex **162** of the ventricular anchors **126** can function like a wedge to facilitate moving the ventricular anchors behind respective leaflets while minimizing the risk of chordae entanglement. The end portions **152**, **154** of each wire **150** can extend downward from attachment locations **156**, **158**, respectively, at the ventricular end **130** of the main body **122**. The wire **150** can then curve back upward from each end portion **152**, **154** toward the apex **162**.

(96) The wires **150** can be covered by biocompatible materials, such as in the embodiment shown in FIGS. **5-7**. A first material **164** can be wrapped around, or coat, at least some portion of the wire **150**. A second material **166** can span across two portions of the wire **150** to form a web, which can improve tissue ingrowth. The first and second materials **164**, **166** can comprise the same material or different materials, such as a biocompatible semi-porous fabric, for example. The covering materials **164**, **166** can increase tissue ingrowth with the ventricular anchor **126** to improve retention. Furthermore, the covering materials can decrease the frictional properties of the ventricular anchors **126** to facilitate implantation and/or increase the frictional properties of the ventricular anchors to improve retention.

(97) FIG. **24** shows exemplary dimensions of the embodiment of the frame **102** shown in FIG. **9**. The diameter “Dmax” of the outer rim **140** of the atrial sealing member **124** can range from about 50 mm to about 70 mm, and is about 50 mm in one example. The diameter “Dbody” of the outer surface of the main body **122** can range from about 23 mm to about 50 mm, and is about 29 mm in one example. The distance “W1” between the two attachment points **156**, **158** for one ventricular anchor **126** can range from about 8 mm to about 50 mm, and is about 25 mm in one example. The overall axial height “Hmax” of the frame **102** can range from about 20 mm to about 40 mm, and is about 30 mm in one example. The axial height “H1” from the outer rim **140** to the lowermost portion **168** of the ventricular anchors **126** can range from about 10 mm to about 40 mm, and is about 23 mm in one example. The axial distance “H2” from the apex **162** of the ventricular anchor **126** to the lowermost portion **168** of the ventricular anchor **126** can range from about 10 mm to about 40 mm, and is about 18 mm in one example. The axial distance “H3” from the lower end **130** of the main body **122** to the lowermost portion **168** of the ventricular anchor **126** can range from about 0 mm to about 10 mm, and is about 5 mm in one example.

(98) Some frame embodiments comprise more than two ventricular anchors. For example, a frame can have two or more ventricular anchors configured to attach to multiple locations along a single leaflet of a native valve. In some such embodiments (not shown), the frame can comprise two ventricular anchors that attach to the anterior mitral leaflet **10** and/or two ventricular anchors that attach to the posterior mitral leaflet **12**. Ventricular anchors can also attach to other regions of the leaflets instead of, or in addition to, the A2 and P2 regions.

(99) Some prosthetic valve embodiments comprise four ventricular anchors spaced evenly apart around a main body. FIGS. **27-32** show one such prosthetic valve embodiment **400** comprising a frame **402** that comprises a pair of ventricular anchors **426** on diametrically opposed sides of a main body **422** and a pair of diametrically opposed commissure anchors **428** located about midway between the ventricular anchors **426**. The ventricular anchors **426** extend downward from attachment points **456** and **458** and comprise a neck portion **450** (see FIG. **31**). These ventricular anchors **426** can function similarly to the ventricular anchors **126** of the frame **102** to capture leaflets and retain the frame **402** within the mitral orifice, as shown in FIG. **33**. The commissure anchors **428** can extend upward from the same attachment locations **456**, **458** on the main body **422** (see FIG. **30**). While the ventricular anchors **426** can clip the mitral leaflets **10**, **12** at the A2 and P2 regions, respectively, the commissure anchors **428** can hook around and extend upward behind the mitral commissures **36**, not compressing the leaflets. The apexes **464** of the commissure anchors **428** can extend upward to abut the ventricular side of the mitral annulus **8** and compress the mitral annulus **8** between the outer rim **440** of the atrial sealing member **424** and the apexes **464** of the commissure anchors **428**. This compression of the mitral annulus **8** can provide

additional retention against both atrial and ventricular movement.

(100) Other frame embodiments can comprise more than four ventricular anchors. For example, a frame can comprise six or more ventricular anchors that can engage multiple locations on the leaflets **10, 12** and/or the commissures **36**.

(101) FIG. **34** shows a frame embodiment **502** that comprises extended ventricular anchors **526** that are configured to extend around the ends of the leaflets **10, 12** and extend upward behind the leaflets to locations proximate the outer rim **540** of a downwardly extending frustoconical atrial sealing member **524**. The upper apexes **562** of the extended ventricular anchors **526** contact the ventricular surface of the mitral annulus **8** and/or portions of the native leaflets **10, 12** adjacent to the annulus, or annulus connection portions of the leaflets, while the outer rim **540** of the atrial sealing member **524** contacts the atrial surface of the mitral annulus and/or the annulus connection portions of the leaflets. The extended ventricular anchors **526** and the atrial sealing member **524** can be configured to oppose one another and desirably compress the mitral annulus **8** and/or annulus connection portions of the leaflets **10, 12** to retain the frame **502** from movement in both the atrial and ventricular directions. Thus, in this embodiment, the ventricular anchors **526** need not compress the native leaflets **10, 12** against the outer surface of the main body **522** of the frame. Instead, as shown in FIG. **34**, the leaflets **10, 12** can be captured loosely between the extended ventricular anchors **526** and the outer surface of the main body **522**.

(102) FIGS. **35** and **36** show a frame embodiment **602** comprising necked, “S” shaped ventricular anchors **626**. From the side view of FIG. **35**, the “S” shape of the ventricular anchors **626** is apparent. Starting from one attachment point A on the ventricular end **630** of the main body **622**, the ventricular anchor wire **650** extends downward and radially outward from the main body to a point B, then curves upward and outward to a point C, then curves upward and inward to a point D, and then curves upward and back outward to an uppermost point, or apex, E. The ventricular anchor wire **650** then continues to extend back to the second attachment point following a similar, but mirrored path. From the frontal view of FIG. **36**, the ventricular anchor wire **650** forms a necked shape that is symmetrical about a longitudinal center axis **690** extending through the center of the main body **622**, forming two mirrored halves. Each half of ventricular anchor wire **650** begins at an attachment point A on the ventricular end **630** of the main body **622**, curves downward and inward (toward the other half) to point B, then curves upward and inward to a necked portion at point C, then curves upward and outward (away from the other half) to a point D, then curves upward and inward again to an uppermost point, or apex, E where the two halves join together. Referring to FIG. **35**, the radial distances from a longitudinal center axis **690** of the main body **622** to points C and E are both greater than the radial distances from the axis **690** to points D. Furthermore, the distance between the two points C is less than the distance between the two points D. The “S” shaped ventricular anchor **626** can help distribute stresses more evenly along the wire **650** and reduce stress concentrations at certain locations, such as the attachment points A.

(103) FIGS. **37** and **38** show a frame embodiment **702** that comprises two wider shaped ventricular anchors **726**. Each wider shaped ventricular anchors **726** comprises a necked mid portion **780** and a broad upper portion **782**. The upper portion **782** can extend generally parallel to the inflow opening of the frame **702** and can be curved around the outer surface of a main body **722**. This wider shape can increase surface contact with the native leaflet and/or other cardiac tissue to reduce pressure and thereby reduce abrasion. In some embodiments, the broad upper portion **782** of the wider shaped ventricular anchors **726** can have a curvature that corresponds to the curvature of the outer surface of the main body **722** (see FIG. **38**) to further improve tissue contact. The wider shaped ventricular anchor can have a longer surface contact with the atrial sealing member; thereby increasing retention performance and reducing paravalvular leak.

(104) FIG. **39** shows a frame embodiment **802** comprising ventricular anchors **826** that are configured to define a separation, or gap, between the anchors and the main body **822** even after the frame **802** expands (although the anchors **826** can otherwise function similar to ventricular anchors **126**, such that the gaps between the anchors **826** and the frame main body **822** can increase and decrease upon compression and expansion of the main body, respectively, to facilitate placement of the anchors **826** behind the native leaflets). The gap can be sized to facilitate capturing the native leaflets **10, 12** with

little or no compression of the native leaflets. Since little or no leaflet compression occurs, this frame embodiment **802** can minimize trauma to the native leaflets. Instead of compressing the leaflets **10**, **12** for valve retention, the ventricular anchors **826** can hook the ventricular edges **40**, **42** of the leaflets **10**, **12**, respectively, while an atrial sealing member **824** of the frame presses downwardly on the atrial side of the mitral valve annulus **8**. The contact between the atrial sealing member **824** and the annulus **8** causes the main body **822** to shift slightly upwardly pulling the ventricular anchors **826** upwardly against the ventricular edges of the leaflets **10**, **12**. The upward force of the ventricular anchors in conjunction with downward tension on the leaflets from the chordae tendineae **16** restrain the implant from moving upward toward the left atrium **4**.

(105) FIG. **40** shows a frame embodiment **902** that comprises a main body **922**, ventricular anchors **926** and a disk-like atrial sealing member **924** that extends radially outward from the upper end **932** of the main body **922**. In this embodiment, the atrial sealing member **924** extends substantially perpendicular to the frame opening defined by the upper and **932** rather than downwardly toward the frame's lower end **930**. The disk-like atrial sealing member **924** can be positioned flat across the top surface of the mitral annulus **8** and provide increased surface area contact for tissue ingrowth.

(106) FIGS. **41** and **42** show frame embodiments **1002** and **1012**, respectively, that comprise an atrial sealing member **1024** having a generally frustoconical portion **1028** that extends from the upper end **1032** of a main body **1022** both radially outward and axially upward away from the main body. The atrial sealing member **1024** can also include a generally cylindrical upper, or inlet, portion **1029** that extends further upward from the frustoconical portion **1028** opposite the upper end **1032** of the main body **1022**. The atrial sealing member **1024** can generally correspond to the shape of the atrial walls **18** adjacent to the mitral annulus **8** and provide for increased contact area between the atrial wall tissue and the atrial sealing member **1024**. The frame **1002** includes ventricular anchors **1026** that extend from a ventricular end **1030** of the main body **1022** and along the majority of the length of the main body.

(107) The frame **1012** shown in FIG. **42** comprises extended ventricular anchors **1050**. The extended anchors **1050** can extend from the ventricular end **1030** of the main body **1022** along the outer surface of the main body and bend radially outward to form upper portions **1060** that extend along the lower surface of the frustoconical portion **1028**. This configuration can allow the extended ventricular anchors **1050** to trap more of the leaflets **10**, **12** and/or the mitral annulus **8** against the frame, thereby reducing paravalvular leakage and improving tissue ingrowth and retention.

(108) FIG. **43** shows a frame embodiment **1102** having ventricular anchors **1126** that have shorter moment arms **D2** compared to the ventricular anchors **126** of the frame **102** shown in FIG. **9**. The shorter moment arms **D2** can result in reduced torque at the ventricular anchor attachment points **1156**, **1158**. The distance **D2** can be reduced by increasing the distance **D1** between the attachment points **1158** and **1156** on the main body **1122** of neighboring ventricular anchors **1126**. The distance **D1** between the ventricular anchors **1126** of the frame **1102** is greater than the distance **D1** between the attachment points **158** and **156** of frame **102** (see FIG. **9**), thus shortening the moment arm **D2** of the force **F** relative to the attachment point **1156**. The reduced torque at the attachment points **1156** and **1158** can reduce fatigue and thus improve the durability of the frame **1102**.

(109) Some embodiments of ventricular anchors can optionally also comprise one or more barbs (not shown) that can protrude radially from a ventricular anchor toward the ventricular walls **20** or toward the leaflets **10**, **12**. Such barbs can help retain a frame, particularly against movement towards the left ventricle **6**.

(110) FIGS. **44A-44D** illustrate a frame embodiment **1202** comprising “flip-up” ventricular anchors **1226**. Each ventricular anchor **1226** can be finger-like and can extend from only one attachment point on the lower end **1230** of the main body **1222**. Alternatively, each ventricular anchor can comprise a wire or similar element that extends from two attachment points on the main body **1222**. In the illustrated embodiment, the ventricular anchors **1226** can be pre-formed to extend along the outer side of the main body **1222** in the functional, deployed state, as shown in FIG. **44D**. During delivery, the ventricular anchors **1226** can be partially or completely straightened, as shown in FIG. **44A**, and retained in that state by a delivery device, such as a sheath. As the frame **1202** is advanced from the sheath, for example, the ventricular anchors **1226** spring back to their pre-formed shape, as shown in



FIGS. 44B-44D, capturing the leaflets **10**, **12** between the ventricular anchors **1226** and the main body **1222**.

(111) FIGS. 45A-45E represent a frame embodiment **1302** comprising “curl-up” ventricular anchors **1326**. As with the ventricular anchors **1226** of FIG. 44, each ventricular anchor **1326** can be finger-like and can extend from two or more points on lower end **1330** of the main body **1322**. The ventricular anchors **1326** can be pre-formed in a curved shape, as shown in FIG. 45E, that extends along the side of the main body **1322** in the deployed state. During delivery, the ventricular anchors **1326** can be partially or completely straightened, as shown FIG. 45A, and retained in that state by a delivery device, such as a sheath. As the frame **1302** is advanced from the sheath, for example, the ventricular anchors **1326** are allowed to spring back to their pre-formed curved shape, as shown in FIGS. 45B-45E, capturing the leaflets **10**, **12** between the ventricular anchors **1326** and the main body **1322**.

(112) In some frame embodiments, one or more ventricular anchor components can be formed separately from the main body and later assembled together to form a frame. In one such frame embodiment **1402**, as shown in FIGS. 46A-46C, a main body **1422** is formed separately from at least one ventricular anchor portion **1424**. The ventricular anchor portions **1424** can comprise one or more ventricular anchors **1426** extending from an at least partially annular base **1432**, which can comprise side-by-side “V” shaped strut portions connected together at their upper ends. The lower ends of the ventricular anchors **1426** in the illustrated embodiment are connected to the base **1432** at the lower vertexes of the “V” shaped portions. After the main body and the ventricular anchor portions are separately formed, the ventricular anchor portions **1424** can be attached to the lower portion **1430** of the main body **1422**. For example, the bases **1432** can be placed on opposite sides of the outer surface of the main body **1422** and then sewn, welded, or otherwise attached to the lower portion **1430** of the main body **1422** in a suitable manner, such as by using a locking mechanism. The bases **1432** can be attached to the main body **1422** such that the “V” shaped portions of the bases overlap with corresponding “V” shaped portions of the lower end **1430** of the main body **1422**. In some embodiments, the ventricular anchor portion **1424** can comprise a complete ring having all of the ventricular anchors **1426** extending from one annular base such that the ventricular anchors are pre-spaced relative to one another. The annular base can then be attached around the lower end **1430** of the main body **1422**. In other embodiments, multiple ventricular anchor portions **1424**, each having one or more ventricular anchors **1426** extending from a respective base **1432** comprising a partial ring, are secured to the main body **1422**.

(113) FIGS. 47A-47D and FIGS. 48A-48C show alternative frame embodiments wherein one or more ventricular anchor components are formed separately from a main body and later assembled together to form a frame. In these frame embodiments, the main body can comprise attachment portions to which anchor portions can be attached using sleeves. For example, FIGS. 47A-47D show an exemplary frame **1500** comprising a main body **1502** having at least two ventricular anchor attachment portions **1508** and at least one ventricular anchor **1504** having two attachment portions **1510** connected to respective attachment portions **1508** with respective sleeves **1506**. Similarly, FIG. 48A-48C show an exemplary frame **1600** comprising a main body **1602** having at least two ventricular anchor attachment portions **1608** and at least one ventricular anchor **1604** having two attachment portions **1610** connected to respective attachment portions **1608** with respective sleeves **1606**. The sleeves can comprise, for example, a metal material, such as Nitinol, having superelastic and/or shape-memory characteristics. In some embodiments, the sleeves can comprise metal of an anneal state suitable for a crimping process. The sleeves can be attached to the anchor portions and to the attachment portions of the main body by any suitable attachment means, such as by welding. As shown in FIGS. 48A-48C, the attachment portion **1610** of the anchors **1604** and the attachment portions **1608** of the main body **1602** can comprise geometric features, such as narrow regions, or cut-outs, which allow the sleeves **1606** to integrate the anchor portions **1604** to the main body **1602**, such as by forming a mechanical lock.

(114) Multi-part construction of a frame, as shown in FIG. 46-48, can reduce strain and fatigue at the ventricular anchor attachment locations compared to a unibody, or one-piece, construction. By contrast, in some embodiments comprising a unibody construction, the ventricular anchors are initially laser cut and expanded such that they extend downward from the lower end of the main body, and are then

formed, or bent, to a desired configuration adjacent to the outside of the main body of the frame. Such bending can strain and weaken the bent portion.

(115) To avoid strain caused by plastic deformation of the ventricular anchors, the ventricular anchors can be pre-formed in a desired implantation (deployed) shape without plastically bending the ventricular anchors. The ventricular anchors can then be elastically deformed, such as straightened and/or compressed, to fit into a delivery device for delivery through the body to the mitral valve region of the heart. The deformed ventricular anchors can resiliently regain their pre-formed shape once freed from the axial constraint of a delivery device to facilitate capturing the leaflets **10, 12** between the ventricular anchors and the main body of the frame.

(116) Any of the various embodiments of frames described above can be combined with a fluid-occluding member, such as valve structure **104**, to form a fully assembled prosthetic valve that can be implanted within the native mitral valve. In other embodiments, any of the frames described above can be provided without a fluid-occluding member and can be used as a scaffolding or docking structure for receiving a separate prosthetic valve in a two-stage delivery process. With reference to the exemplary embodiment shown in FIG. **70**, a docking frame **103** (which can have a construction similar to the frame **102**) can be deployed first, for example by any of the anchoring techniques discussed above. Then, a separate prosthetic valve **114** can be delivered and deployed within the lumen formed by the previously deployed docking frame **103**. The separate prosthetic valve **114** desirably comprises a radially compressible and expandable frame **116** that mounts a fluid-occluding member (not shown in FIG. **70**), such as the valve structure **104** (see FIG. **7**) having a plurality of leaflets **106**. When expanded inside the docking frame **103**, the frame **116** of the prosthetic valve **114** engages the inside surface of the docking frame **103** so as to retain, such by friction or mechanical locking feature, the prosthetic valve **114** within the docking frame **103**. Examples of prosthetic valves that can be used in such a two-stage process are disclosed in U.S. Pat. No. 7,510,575, which is incorporated herein by reference. In particular embodiments, the prosthetic valve can comprise any of various transcatheter heart valves, such as the Sapien valve, available from Edwards Lifesciences LLC (Irvine, CA).

(117) The technique of capturing the leaflets **10, 12** between a ventricular anchor and the main body of a frame, such as shown in FIG. **23**, can provide several advantages. First, this can allow for anchoring onto the native leaflets **10, 12** for retention within the mitral valve region. Second, this technique can utilize the native chordae **16** for retention. Third, this technique can prevent the anterior leaflet **10** from being “pulled” toward the aortic valve **14** when the left ventricle **6** contracts and blood rushes out through the aortic valve (systolic anterior motion). Fourth, this technique tends to force the native leaflets **10, 12** to collapse around the main body of the frame, which can reduce leakage between the outside of the prosthetic valve **100** and the native mitral valve **2**. Fifth, this technique allows for implantation from either the left atrium **4** or from the left ventricle **6**, as described in detail below.

(118) As described above, various frame embodiments can utilize one or more anchoring techniques other than compressing the leaflets **10, 12** to retain the prosthetic valve **100** in a desired position within the mitral valve orifice. These anchoring techniques can include, for example, utilizing tension of the native chordae **16**, extending the ventricular anchor length such that the apex of the ventricular anchor is pressed up against the mitral annulus **8** so as to form a stop, and compressing the mitral annulus **8** and/or atrial tissue between the apex of an ventricular anchor and the outer rim of an atrial sealing member of the frame.

#### (119) Delivery Approaches

(120) The various methods and apparatus described hereinafter for delivery and implantation at the native mitral valve region are described with respect to the prosthetic valve **100**, though it should be understood that similar methods and apparatus can be used to deliver and/or implant a component of the prosthetic valve **100**, such as the frame **102** without the valve structure **104**, or other prosthetic apparatus.

(121) The prosthetic valve **100** can be delivered to the mitral valve region from the left ventricle **6** or from the left atrium **4**. Because of the anatomy of the native mitral valve **2**, different techniques and/or equipment can be used depending on the direction the prosthetic valve **100** is delivered.

(122) Delivery from the ventricular side of the mitral annulus **8** can be accomplished in various

manners. For example, the prosthetic valve **100** can be delivered via a transapical approach in which access is made to the left ventricle **6** via the heart apex **38**, as shown in FIG. 57.

(123) Delivery from the atrial side of the mitral annulus **8** can also be accomplished in various manners. For example, a transatrial approach can be made through an atrial wall **18**, as shown in FIG. 66, for example by an incision through the chest. An atrial delivery can also be made from a pulmonary vein **32** (see FIG. 1). In addition, atrial delivery can be made via a transeptal approach, as shown in FIG. 67, wherein an incision is made in the atrial portion of the septum **30** to allow access from the right atrium **26**, such as via the inferior or superior vena cava **34**.

#### (124) Ventricular Approaches

(125) One technique for delivering a compressed prosthetic apparatus, such as the prosthetic valve **100**, to the mitral valve region includes accessing the native mitral valve region from the left ventricle **6**, one example being the transapical approach. Alternatively, access to the left ventricle **6** can be made through the aortic valve **14**. In the transapical approach, access to the left ventricle **6** can be made through an incision in the chest and an incision at the heart apex **38**, as shown in FIG. 57. A transapical delivery system can be used with the transapical approach.

(126) FIGS. 49-53 show an exemplary transapical delivery system, or delivery tool, **2000** that is configured to deliver and implant the prosthetic valve **100**. The delivery system **2000** can comprise a series of concentric shafts and sheaths aligned about a central axis and slidable relative to one another in the axial directions. The delivery system **2000** can comprise a proximal handle portion **2002** for physician manipulation outside of the body while a distal end portion, or insertion portion, **2004** is inserted into the body.

(127) The delivery system **2000** can comprise an inner shaft **2006** that runs the length of the delivery system and comprises a lumen **2008** through which a guidewire (not shown) can pass. The inner shaft **2006** can be positioned within a lumen of a pusher shaft **2010** and can have a length that extends proximally beyond the proximal end of the pusher shaft and distally beyond the distal end of the pusher shaft. The delivery system **2000** can comprise an annular space **2012** between the outer surface of the inner shaft **2006** and the inner surface of the pusher shaft **2010**. This annular space can be used for flushing with saline or for allowing blood to be expelled distally.

(128) The delivery system **2000** further comprises an inner sheath **2014** positioned concentrically around at least a distal portion of the pusher shaft **2010**. The inner sheath **2014** is axially slidable relative to the pusher shaft **2010** between a delivery position (see FIG. 55) and a retracted position (see FIG. 50). In the delivery position, a distal end portion **2016** of the inner sheath **2014** is positioned distal to a distal end, or pusher tip **2018**, of the pusher shaft **2010**. In the delivery position, the distal end portion **2016** of the inner sheath **2014** forms an inner cavity that can contain a compressed prosthetic valve **100**. In the retracted position (see FIG. 50), the distal end **2017** of the inner sheath **2014** is positioned proximal to or aligned axially with the pusher tip **2018**. As the inner sheath **2014** moves from the delivery position toward the retracted position (either by retracting the inner sheath **2014** proximally relative to the pusher shaft **2010** or advancing the pusher shaft distally relative to the inner sheath), the pusher tip **2018** can force the prosthetic valve **100** out of the distal end portion **2016** of the inner sheath.

(129) As shown in FIG. 50, the inner sheath **2014** comprises one or more longitudinally disposed slots **2028** extending proximally from a distal end **2017** of the inner sheath. These slots **2028** can allow ventricular anchors **126** of a prosthetic valve **100** contained within the inner sheath **2014** to extend radially outward from the compressed main body of the prosthetic valve while the main body is retained in the compressed state within the inner sheath. In the embodiment shown in FIG. 50, two slots **2028** are shown oriented on diametrically opposed sides of a longitudinal central axis of the inner sheath **2014**. This embodiment corresponds to the prosthetic valve **100**, which comprises two opposed ventricular anchors **126**. In other embodiments, the inner sheath **2014** can comprise a different number of slots **2028**, for example four slots, that correspond to the number and location of ventricular anchors on a selected prosthetic valve. In some embodiments, such as shown in FIG. 50, the proximal end portion **2020** of the each slot **2028** comprises a rounded opening that has a greater angular width than the rest of the slot.

(130) A break-away, or frangible, retaining band **2022** can be positioned around the distal end portion

2016 of the inner sheath 2014, as shown in FIG. 50. The band 2022 can help retain the distal end portion 2016 of the inner sheath 2014 from splaying apart from the force of a compressed prosthetic valve 100 contained within the inner sheath 2014. The band 2022 comprises a proximal edge 2024 that can comprise at least one notch 2026 located over a slot 2028 in the inner sheath 2014. The band 2022 can comprise a frangible material and can be configured to tear or break apart at the notch location when a sufficient axial force is applied at the notch 2026. In use, the band 2022 is configured to break at notches 2026 under the force of the ventricular anchors 126 of the valve 100 as it is deployed from the inner sheath 2014, as further described below.

(131) An outer sheath 2036 is positioned concentrically around a portion of the inner sheath 2014 and is slidable axially relative to the inner sheath. The outer sheath 2036 can be positioned to cover at least a portion of the distal end portion 2016 of the inner sheath 2014. In such a covered position, such as shown in FIG. 55, the ventricular anchors can be contained between the inner and outer sheath. The outer sheath 2036 is in this covered position while the loaded delivery system 2000 is inserted through the body and into the left ventricle 6. The outer sheath 2036 can be retracted proximally relative to the sheath 2014 to uncover the slots 2028 and allow the ventricular anchors 126 to spring outward through the slots in the inner sheath 2014 during deployment. Alternatively, the inner sheath 2014 can be advanced distally relative to the outer sheath 2036 to uncover the slots 2028.

(132) With reference to FIG. 51, the handle portion 2002 of the delivery system 2000 can comprise components that facilitate sliding the inner sheath 2014 and the outer sheath 2036 back and forth along their respective ranges of axial movement to load, deliver, and deploy the prosthetic valve 100. An outer sheath grip 2052 can be attached to the proximal end of the outer sheath 2036. A physician can grasp the outer sheath grip 2052 and push or pull the outer sheath 2036 proximally or distally relative to the rest of the delivery system 2000. The outer sheath can also be mounted on a lead screw (not shown). The handle portion 2002 of the delivery system 2000 can further comprise a housing 2054 that provides a hand grip or handle for the physician to hold the delivery system 2000 steady while she uses the other hand to actuate the sheaths. A sliding lead screw 2056 can be fixed (e.g., bonded, mechanically locked, etc.) to a proximal end portion 2058 of the inner sheath 2014 and be positioned within the housing 2054. The lead screw 2056 can be fixed rotationally relative to the housing 2054 and can be constrained to an axial sliding range within the housing. A rotatable sleeve 2060 can be positioned concentrically between the outer housing 2054 and the inner lead screw 2056 and can comprise a proximal knob portion 2062 that extends free of the housing 2054 to provide a hand grip for the physician to rotate the rotatable sleeve 2060. The rotatable sleeve 2060 can be free to rotate relative to the housing 2054, but be fixed axially relative to the housing. The lead screw 2056 can comprise an outer helical groove 2064 that interacts with inwardly projecting ridges 2066 on the rotatable sleeve 2060 such that when the knob 2062 is rotated relative to the lead screw 2056 and the housing 2054, the ridges 2066 cause the lead screw 2056 to slide axially, thereby causing the inner sheath 2014 to also slide axially. Thus, the physician can move the inner sheath 2014 proximally by rotating the knob 2062 one direction relative to the housing 2054 and distally by rotating the knob the opposite direction relative to the housing. The housing 2054 can be fixed relative to the pusher shaft 2010 such that when the knob 2062 is rotated relative to the housing, the lead screw 2056 and the inner sheath 2014 slide axially together relative to the pusher shaft 2010 and the housing 2054.

(133) As shown in FIG. 51, the inner shaft 2006 passes all the way through the handle portion 2002 of the delivery system 2000 and the pusher shaft 2010 can terminate at or near a proximal end cap 2068 of the handle portion 2002. The annular space 2012 between the outer surface of the inner shaft 2006 and the inner surface of the pusher shaft 2010 (see FIGS. 52 and 53) can be fluidly connected to at least one flushing port 2070 in the end cap 2068 of the handle portion 2002. The flushing port 2070 can provide access to inject fluid into the annular space 2012 and/or allow fluid to escape from the annular space.

(134) As shown in FIG. 49, a nose cone 2030 can be attached to the distal end of the inner shaft 2006. The nose cone 2030 can be tapered from a proximal base 2034 to a distal apex 2032. The base 2034 can have a diameter about equal to the diameter of the outer sheath 2036. The nose cone 2030 can be retracted proximally, by sliding the inner shaft 2006 proximally relative to the rest of the delivery system 2000, to mate against the distal end of the outer sheath 2036 and/or the inner sheath 2014 to

prosthesis contain the compressed prosthetic valve **100**, as shown in FIG. 55. The nose cone **2030** can also be moved distally away from the sheaths to provide space for the prosthetic valve **100** to be loaded and/or deployed. During insertion of the delivery system **2000** through the body, the tapered nose cone **2030** can act as a wedge to guide the insertion portion **2004** of the delivery system **2000** into the body and provides an atraumatic tip to minimize trauma to surrounding tissue as the delivery system is advanced through the body.

(135) To load the prosthetic valve **100** into the delivery system **2000**, the nose cone **2030** must be moved distally away from the sheaths and the inner sheath **2014** must be advanced distally to the delivery position, as shown in FIG. 54 (without retaining band **2022**). The outer sheath **2036** can be retracted to expose the slots **2028** in the inner sheath **2014**. The prosthetic valve **100** is then positioned between the nose cone **2030** and the inner sheath **2014** and around the inner shaft **2006**. The prosthetic valve **100** is then compressed to the compressed state and slid into the inner sheath **2014** such that the proximal, or lower, end of the prosthetic valve is adjacent to or contacting the pusher tip, as shown in FIG. 56. A loading cone or equivalent mechanism can be used to insert the valve **100** into the inner sheath **2014**. In embodiments of the prosthetic valve **100** comprising a pusher member **204**, such as in FIG. 25, the bottom end **206** of the pusher member **204** can contact the pusher tip **2018**, as shown in FIG. 56. The ventricular anchors **126** can be allowed to extend out through the rounded proximal end portions **2020** of the respective slots **2028**, as shown in FIG. 54. The proximal end portion **2020** of each slot can have sufficient angular width to allow the two end portions of the ventricular anchor **126** to reside side-by-side within the slot, which can cause the intermediate portion of the ventricular anchor to assume a desired shape for implanting behind the leaflets **10**, **12**. The break-away retaining band **2022** can be placed around the distal end portion of the inner sheath **2014** such that each notch **2026** in the band **2022** is located over a respective slot, as shown in FIG. 50. The outer sheath **2036** is then advanced distally to cover the slots **2028**, as shown in FIG. 55, thereby compressing the ventricular anchors **126** and constraining the ventricular anchors within the outer sheath **2036**. Alternatively, the prosthetic valve can be inserted into the inner sheath **2014** while the outer sheath **2036** is covering the slots **2028**, such that the ventricular anchors **126** are positioned in the slots, but cannot extend out of the slots. The ventricular anchors **126** can also be constrained between the outer surface of the inner sheath **2014** and inner surface of the outer sheath **2036**. In any case, the ventricular anchors **126** are free to spring radially outward once the outer sheath **2036** is retracted. After the prosthetic valve **100** is within the inner sheath **2014**, the inner shaft **2006** can be retracted to pull the nose cone **2030** against the distal end of the inner sheath **2014** and/or the outer sheath **2036**, as shown in FIG. 55. With the prosthetic valve **100** within the inner shaft **2006**, the nose cone **2030** retracted and the outer sheath **2036** constraining the ventricular anchors **126**, the delivery system **2000** is in the loaded configuration and ready for insertion into the body.

(136) In the loaded configuration shown in FIG. 55, the loaded delivery system **2000** can be inserted, nose cone **2030** first, through heart apex **38** into the left ventricle **6** and positioned near the mitral valve region for deployment. An introducer sheath (not shown) can be initially inserted through an incision in the heart to provide a port for introducing the delivery system **2000** into the heart. In addition, the delivery system **2000** can be advanced over a conventional guide wire (not shown) that is advanced into the heart ahead of the delivery system **2000**. The grip **2052** can then be moved proximally relative to the rest of the delivery system to retract the outer sheath **2036** relative to the inner sheath **2014** and allow the ventricular anchors **126** to spring outwardly away from the inner sheath **2014**, as shown in FIGS. 56 and 57, such that the ventricular anchors extend through the rounded proximal end portion **2020** of the slots **2028**. The delivery system desirably is oriented rotationally such that each ventricular anchor **126** is positioned between sets of chordate tendineae **16** attached to one of the native mitral valve leaflets **10**, **12**. Next, the delivery system **2000** can be advanced atrially such that the nose cone **2030** enters the native mitral valve orifice and the protruding ventricular anchors **126** move between respective leaflets **10**, **12** and the ventricular walls **20**, as shown in FIG. 58. Then, while holding a housing **2054** of the delivery system **2000** steady, the physician can rotate the knob **2062** of the rotatable sleeve **2060** relative to the housing to retract the inner sheath **2014** proximally. The pusher tip **2018** remains stationary while the inner sheath **2014** retracts, thereby leaving the compressed prosthetic valve **100** in the same axial

location as it is uncovered and deployed from the inner sheath **2014**. Alternatively, the inner sheath **2014** can be held stationary while the pusher tip **2060** is moved distally to push the valve **100** out of the inner sheath **2014**. While the inner sheath **2014** is being retracted relative to the pusher tip **2018**, the pusher tip can exert an axial force in the distal direction upon the proximal, or lowermost, surface of the prosthetic valve **100**. In embodiments of the prosthetic valve having a pusher member **204**, the pusher member **204** can direct this axial force directly to the main body **122** and prevent direct contact between the pusher tip **2018** and the ventricular anchor **126** to reduce the risk of damage to the ventricular anchors.

(137) When the inner sheath **2014** is retracted relative to the prosthetic valve **100**, the distal, or upper, portion of the prosthetic valve comprising the downwardly folded atrial sealing member **124** is uncovered first. With reference to FIGS. **59** and **60**, when the inner sheath **2014** has been retracted beyond the outer rim of the atrial sealing member **124** of the prosthetic valve **100**, the atrial sealing member can spring radially outward away from the main body **122**, pivoting about the distal end of the main body.

(138) As the inner sheath **2014** is retracted relative to the prosthetic valve **100**, the end portions of the ventricular anchors **126** passing through the rounded proximal end portion **2020** of the slots **2028** are forced through the narrower distal portions of the slots **2028** toward the retaining band **2022**, as shown in FIGS. **59** and **60**. The end portions of the ventricular anchors are initially side-by-side in the wider proximal end portion **2020** of the slot. When forced into the narrower portion of a slot **2028**, the two end portions of each ventricular anchor **126** can be radially overlapping, or oriented one on top of the other, as opposed to side-by-side. In other embodiments, the slots **2028** can be wider such that the two end portions of the ventricular anchor **126** can move about the slots **2028** side-by-side. As the ventricular anchor **126** moves toward the distal end of a slot **2028**, the ventricular anchor can contact the notch **2026** in the retaining band **2022**, as shown in FIG. **60**, and can cut the band **2022** or otherwise cause the band to tear or split apart at the notched location, as shown in FIG. **61**. When the inner sheath **2014** is retracted beyond the proximal, or lower, end of the prosthetic valve **100**, the compressed body of the prosthetic valve can resiliently self-expand to the expanded state, as shown in FIG. **61**. As the prosthetic valve expands, the gaps between the ventricular anchors **126** and the outer surface of the main body **122** decreases, capturing the leaflets **10**, **12** between the ventricular anchors **126** and the main body **122**, as shown in FIGS. **23** and **62**. The expansion of the main body **122** of the prosthetic valve **100** can force open the native mitral leaflets **10**, **12**, holding the native mitral valve **2** in an open position. The prosthetic valve **100** can then replace the functionality of the native mitral valve **2**. After the prosthetic valve **100** is expanded, the inner shaft **2006** of the delivery system can be retracted, pulling the nose cone **2030** back through the prosthetic valve, and the whole delivery system **2000** can be retracted out of the body.

(139) In some embodiments, the delivery system **2000** can be guided in and/or out of the body using a guide wire (not shown). The guide wire can be inserted into the heart and through the native mitral orifice, and then a proximal end of the guidewire can be threaded through the lumen **2008** of the inner shaft **2006**. The delivery system **2000** can then be inserted through the body using the guidewire to direct the path of the delivery system.

#### (140) Atrial Approaches

(141) The prosthetic valve **100** can alternatively be delivered to the native mitral valve region from the left atrium **4**. Referring to FIGS. **63-67**, one approach for delivering the prosthetic valve from the atrial side of the mitral valve region utilizes a delivery catheter **2100**. The prosthetic valve **100** is first crimped from the expanded state to the radially compressed state and loaded into a primary sheath **2102**, and optionally also a secondary sheath, at the distal end portion of the delivery catheter **2100**, as shown in FIG. **63**. The delivery catheter **2100** is used to guide the prosthetic valve **100** through the body and into the left atrium **4**. The prosthetic valve **100** is oriented within the sheath **2102** such that the outflow end **112** of the prosthetic valve **100** (the end supporting the ventricular anchors **126**) is closest to the distal end of the sheath and thus enters the left atrium **4** first and the inflow end **110** (the atrial sealing member **124**) of the prosthetic valve enters last. The sheath **2102** can then be inserted into the left atrium **4** in various manners, one example being the transatrial approach shown in FIG. **66**, and another example being the transeptal approach shown in FIG. **67**. When the delivery catheter **2100** is used to access the

heart via the patient's vasculature, such as shown in FIG. 67, the catheter **2100** can comprise a flexible, steerable catheter.

(142) Once in the left atrium **4**, the distal end **2104** of the primary sheath **2102** can be moved across the mitral annulus **8** such that the ventricular anchors **126** are positioned beyond the mitral leaflets **10**, **12** prior to deploying the ventricular anchors from the sheath.

(143) The prosthetic valve **100** can then be partially expelled from of the distal end **2104** of the primary sheath **2102** using a rigid pusher shaft **2106** (see FIG. 64) that is positioned within the sheath **2102** and can slide axially relative to the sheath. When the sheath **2102** is retracted proximally relative to the pusher shaft **2106** and the prosthetic valve **100**, the pusher shaft **2106** urges the prosthetic valve distally out of the sheath **2102**, as shown in FIG. 64. Alternatively, the pusher shaft **2106** can be moved distally while the sheath **2102** is held in place, thereby pushing the prosthetic valve **100** distally out of the sheath.

(144) When the primary sheath **2102** is inserted across the mitral annulus **8** and past the lower ends of the leaflets **10**, **12**, the prosthetic valve **100** can be partially expelled to free the ventricular anchors **126**, as shown in FIG. 64. The freed ventricular anchors **126** can spring outwardly when they are freed from the sheath **2102**. Optionally, the sheath **2102** can then be slid back over the exposed portion of the main body **122**, such that only the ventricular anchors are showing, as shown in FIG. 65. To accomplish this step, the atrial end of the frame can comprise features (not shown), such as mechanical locking features, for releasably attaching the prosthetic valve **100** to the pusher shaft **2106**, such that the pusher shaft can pull the prosthetic valve back into the sheath **2102**. The sheath **2102** and the prosthetic valve **100** are then retracted atrially, proximally, such that the outwardly protruding ventricular anchors **126** move between respective leaflets **10**, **12**, and the ventricular walls **20**, as shown in FIGS. 66-68. In other embodiments, such as those shown in FIGS. 44 and 45, the ventricular anchors can elastically deflect upward or bend around respective leaflets **10**, **12** when the ventricular anchors are freed from the sheath **2102**.

(145) Optionally, the delivery catheter **2100** can also include a secondary sheath (not shown) within the outer sheath **2102** and can contain the pusher shaft **2106**, the atrial sealing member **124** and the main body **122** of the frame, but not the anchors **126**. In the position shown in FIG. 63, the distal end of the secondary sheath can be positioned between the anchors **126** and the main body **122**. As the outer primary sheath **2102** is retracted, as in FIG. 64, the secondary sheath can remain in a position compressing the main body **122** of the frame while the anchors **126** are freed to extend outward. Because the secondary sheath remains covering and compressing the main body **122**, there is no need recover the main body with the primary sheath **2102**, as in FIG. 65. Instead, the prosthetic valve **100** is moved proximally by moving the secondary sheath and pusher shaft proximally in unison. Then, to expel the prosthetic valve **100** from the secondary sheath, the secondary sheath is retracted proximally relative to the pusher shaft **2106**.

(146) After the ventricular anchors **126** are positioned behind the leaflets **10**, **12** and the remaining portion of the prosthetic valve **100** is expelled from the primary sheath **2102**, the prosthetic valve **100** can expand to its functional size, as shown in FIGS. 62 and 69, thereby capturing the leaflets **10**, **12** between the ventricular anchors **126** and the main body **122**. Once the prosthetic valve **100** is implanted, the delivery catheter **2100** can be retracted back out of the body.

(147) In alternative prosthetic valve embodiments, the main body and the atrial sealing member of the frame can be plastically expandable and can be expanded by a balloon of a balloon catheter (not shown) when the prosthetic valve is positioned at the desired location. The ventricular anchors in such an embodiment can exhibit a desired amount of elasticity to assist in positioning the leaflets **10**, **12** between the ventricular anchors and the main body during deployment. Once the prosthetic valve is fully expanded, the balloon can be retracted through the expanded prosthetic valve and out of the body.

(148) Mitral Regurgitation Reduction

(149) Mitral regurgitation (MR) occurs when the native mitral valve fails to close properly and blood flows into the left atrium from the left ventricle during the systole phase of heart contraction. MR is the most common form of valvular heart disease. MR has different causes, such as leaflet prolapse, dysfunctional papillary muscles and/or stretching of the mitral valve annulus resulting from dilation of

the left ventricle. MR at a central portion of the leaflets can be referred to as central jet MR and MR nearer to one commissure of the leaflets can be referred to as eccentric jet MR.

(150) Rather than completely replacing the native mitral valve, another way to treat MR is by positioning a prosthetic spacer between the leaflets that decreases the regurgitant orifice area, allowing the mitral valve to function with little or no regurgitation, while minimizing impact to the native valve and left ventricle function and to the surrounding tissue. Additional information regarding treatment of MR can be found in U.S. Pat. No. 7,704,277 and U.S. Publication No. 2006/0241745 A1, both of which are incorporated by reference herein.

(151) FIG. 71 shows an exemplary prosthetic spacer embodiment **3000** with which a spacer, or other body, can be suspended or “floated” between the leaflets using anchoring concepts described herein. The prosthetic spacer **3000** can comprise a frame **3002** and spacer body **3004**. The spacer body **3004** can comprise polyurethane, foam, and/or other suitable material(s) and can optionally be coated with Teflon and/or other suitable material(s). The spacer body **3004** can comprise a crescent shape that conforms to the crescent shaped juncture between the anterior leaflet **10** and the posterior leaflet **12** (see FIGS. 4A and 4B), or the spacer body can comprise other suitable shapes, such as an ellipse, circle, hourglass, etc. Depending on the shape of the spacer body **3004** and the positioning of the spacer body relative to the native structure, embodiments of the prosthetic spacer **3000** can help treat central jet MR, eccentric jet MR, or both.

(152) Furthermore, the spacer body **3004** can comprise a minimal transverse cross-sectional area and tapered edges. This shape can reduce diastolic forces from blood flowing through the mitral valve from the left atrium to the left ventricle. This shape can also reduce systolic forces on the spacer body **3004** when the native valve is closed around the spacer body and naturally place a larger portion of the systolic forces on the native leaflets and chordae. The shape of the spacer body **3004** can therefore reduce the forces transferred to the native valve tissue at anchor engagement locations, which can reduce the likelihood of perforation and erosion at the engagement locations and rupture of the native chordae that support the leaflets. The overall minimal size of the prosthetic spacer **3000** can further provide an opportunity to decrease the required cross-sectional size of a delivery system, allowing for delivery via narrower vasculature and/or less invasive incisions in the body and heart.

(153) The frame **3002** can be made of a strong, flexible material, such as Nitinol. As shown in FIG. 71, the frame **3002** can comprise a frame body **3006**, an anterior ventricular anchor **3008**, a posterior ventricular anchor **3010**, an anterior atrial anchor **3012** and a posterior atrial anchor **3014**. The frame body **3006** can comprise a generally longitudinal column extending through the spacer body **3004**. Various embodiments of the frame body **3006** are described in detail below.

(154) The frame **3002** can further comprise one or more spacer expanders **3024** extending laterally from the frame body **3006** through the spacer body **3004**. The expanders **3024** can resiliently expand away from the frame body and assist in the expansion of the spacer body **3004** during deployment. In some embodiments, the spacer expanders **3024** can be rectangular cut-out portions of a cylindrical frame body **3006**, as shown in FIG. 71, that are bent radially away from the frame body.

(155) The anterior ventricular anchor **3008** is configured to extend from the ventricular end of the frame body **3006**, around the A2 edge of the anterior leaflet **10** and extend upward behind the leaflet to a location on the ventricular surface of the mitral annulus **8** and/or the annulus connection portion of the anterior leaflet, while the anterior atrial anchor **3012** is configured to extend radially from the atrial end of the frame body **3006** to a location on the atrial surface of the mitral annulus **8** opposite the anterior ventricular anchor **3008**. Similarly, the posterior ventricular anchor **3010** is configured to extend from the ventricular end of the frame body **3006**, around the P2 edge of the posterior leaflet **12** and extend upward behind the leaflet to a location on the ventricular surface of the mitral annulus **8** and/or the annulus connection portion of the posterior leaflet, while the posterior atrial anchor **3014** is configured to extend radially from the atrial end of the frame body **3006** to a location on the atrial surface of the mitral annulus **8** opposite the posterior ventricular anchor **3010**.

(156) The ventricular anchors **3008**, **3010** and the atrial anchors **3012**, **3014** can comprise broad engagement portions **3016**, **3018**, **3020** and **3022**, respectively, that can be configured to compress the mitral annulus **8** and/or annulus connection portions of the leaflets **10**, **12** to retain the prosthetic spacer



**3000** from movement in both the atrial and ventricular directions. The broad engagement portions can provide a greater area of contact between the anchors and the native tissue to distribute the load and reduce the likelihood of damaging the native tissue, such as perforation or erosion at the engagement location. The ventricular anchors **3008**, **3010** in the illustrated configuration loop around the native leaflets **10**, **12** and do not compress the native leaflets against the outer surface of the spacer body **3004**, allowing the native leaflets to naturally open and close around the spacer body **3004**.

(157) As shown in FIG. **74**, the mitral annulus **8** is generally kidney shaped such that the anterior-posterior dimension is referred to as the minor dimension of the annulus. Because the prosthetic spacer **3000** can anchor at the anterior and posterior regions of the native mitral valve **2**, the prosthetic spacer can be sized according to the minor dimension of the annulus **8**. Echo and CT measuring of the minor dimension of the mitral annulus **8** are exemplary methods of sizing the prosthetic spacer **3000**.

(158) FIGS. **75-79** illustrate an exemplary method for delivering the prosthetic spacer **3000** to the native mitral valve region of the heart. The prosthetic spacer **3000** can be delivered into the heart using a delivery system comprising an outer sheath **3030** and inner torque shaft **3032**. The prosthetic spacer **3000** is compressed and loaded into a distal end of the outer sheath **3030** with the atrial anchors **3012**, **3014** loaded first. As shown in FIG. **75**, the atrial anchors are resiliently extended proximally and the ventricular anchors **3008**, **3010** are resiliently extended distally such that the prosthetic spacer **3000** assumes a sufficiently narrow cross-sectional area to fit within the lumen of the outer sheath **3030**.

Within the outer sheath **3030**, the prosthetic spacer **3000** is positioned such that the atrial end of the frame body **3006** abuts the distal end of the torque shaft **3032**, the atrial anchors **3012**, **3014** are between the torque shaft and the inner wall of the outer shaft, the compressed spacer **3004** abuts the inner wall of the outer sheath, and the distal ends of the ventricular anchors **3008**, **3010** are adjacent to the distal opening of the outer sheath. The torque shaft **3032** can be releasably coupled to the atrial end of the prosthetic spacer **3000**, such as at the proximal end of the frame body **3006**.

(159) Once loaded, the delivery system can be introduced into the left atrium **4**, such as via the atrial septum **30**, and the distal end of the outer sheath **3030** can be passed through the native mitral valve **2** and into the left ventricle **6**, as shown in FIG. **75**.

(160) Next, the outer sheath **3030** can be retracted relative to the torque shaft **3032** to expel the ventricular anchors **3008**, **3010** from the distal opening of the outer sheath. At this point, the torque shaft **3032** can be rotated to rotate the prosthetic spacer **3000** within the outer sheath **3030** (or optionally, the torque shaft and the outer sheath can both be rotated) as needed to align the ventricular anchors with the A2/P2 aspects of the native valve **2**. The releasable attachment between the torque shaft **3032** and the prosthetic spacer **3000** can be sufficient to transfer torque from the torque shaft to the prosthetic in order to rotate the prosthetic as needed. The ventricular anchors **3008**, **3010** can be pre-formed such that, as they are gradually expelled from the outer sheath **3030**, they begin to curl apart from each other and around the A2/P2 regions of the leaflets. This curling movement can be desirable to avoid entanglement with the ventricular walls. When the outer sheath **3030** is retracted to the ventricular end of the frame body **3006**, as shown in FIG. **76**, the ventricular anchors **3008**, **3010** are fully expelled from the outer sheath and positioned behind the leaflets. The entire delivery system and prosthetic can then be moved proximally until the engagement portions **3016**, **3018** of the ventricular anchors abut the ventricular side of the mitral annulus **8** and/or the annulus connection portions of the leaflets **10**, **12**.

(161) Next, the outer sheath **3030** can be further retracted relative to the torque shaft **3032** such that the distal end of the outer sheath is even with the atrial end of the frame body **3006**, as shown in FIG. **77**, which allows the compressed spacer expanders **3024** and the compressed spacer body, or other body, **3004** to resiliently self-expand radially outward to the fully expanded, functional state. Note that the spacer body **3004** expands mostly in a direction perpendicular to the minor dimension of the annulus, or toward the commissures **36** (see FIG. **74**). In some embodiments, the spacer body **3004** can unfold or unfurl from the compressed state to the expanded state and in some embodiments the spacer body can be inflated, such as with saline or with an epoxy that hardens over time.

(162) Once the spacer body is expanded within the valve, as shown in FIG. **77**, hemodynamic evaluation of the spacer can be performed to assess the effectiveness of the prosthetic spacer **3000** in reducing MR. Depending on the result of the evaluation, deployment can continue or the prosthetic

spacer **3000** can be recovered, retracted and/or repositioned for deployment.

(163) From the position shown in FIG. 77, the outer sheath **3030** can be advanced back over the spacer body **3004** (by advancing the outer sheath **3030** relative to the torque shaft **3032**), causing the spacer body to re-compress, as shown in FIG. 76. In some embodiments, the ventricular anchors are not recoverable, though in some embodiments the ventricular anchors can be sufficiently pliable to be re-straightened and recovered, in which case then entire delivery process can be reversed and restarted. From the position shown in FIG. 76, the delivery system can be repositioned and the spacer body **3004** can be redeployed and reassessed.

(164) Once the ventricular anchors **3008**, **3010** and the spacer body **3004** are acceptably deployed, the outer sheath **3030** can be further retracted relative to the prosthetic spacer **3000** and the torque shaft **3032** to expel the atrial anchors **3012**, **3014** from the outer sheath, as shown in FIG. 78. Once fully expelled, the atrial anchors resiliently curl into their final deployment position shown in FIG. 78 with their engagement portions **3020**, **3022** pressed against the atrial side of the annulus **8** and/or the annulus connection portions of the leaflets **10**, **12** opposite the engagement portions **3016**, **3018**, respectively, of the ventricular anchors, thereby compressing the annulus and/or the annulus connection portions of the leaflets at the A2 and P2 regions to retain the prosthetic spacer **3000** within the native mitral valve region **2**.

(165) Once the atrial anchors **3012**, **3014** are deployed, the torque shaft **3032** can be released from the atrial end of the frame body **3006**. The delivery system can then be retracted back out of the body, leaving the prosthetic spacer **3000** implanted, as shown in FIG. 79.

(166) In some embodiments, the spacer body **3004** can comprise a valve structure **3040**, such the embodiments shown in FIGS. 80 and 82. The valve structure **3040** can function in conjunction with the native mitral valve **2** to regulate blood flow between the left atrium **4** and the left ventricle **6**. For example, the valve structure **3040** can be positioned between the native leaflets such that the native leaflets close around the outside of the valve structure such that some blood flows through the valve structure while other blood flows between the outside of the valve structure and the native leaflets. The valve structure **3040** can comprise a three-leaflet configuration, such as is described herein with reference to the valve structure **104** and shown in FIGS. 5-7.

(167) In some embodiments, the frame body **3006** can comprise a cylinder, which can optionally comprise solid-walled tube, such as in FIGS. 71-74, a mesh-like wire lattice **3046**, such as in FIG. 82, or other cylindrical configurations. With reference to FIGS. 71-75, the frame body **3006** and optionally one or more of the anchors can be formed from a solid-walled tube, such as of Nitinol, wherein the atrial anchors are formed, such as by laser cutting, from one portion of the tube and the ventricular anchors are formed from a second portion of the tube and the frame body is formed from a portion of the tube between the first and second portions. The anchors can then be formed, such as by heat treatment, to a desired implantation configuration. In such embodiments, the anchors and the frame body can be a unibody, or monolithic, structure.

(168) In other embodiments, the frame body **3006** can comprise a spring-like helically coiled wire column **3050**, as shown in FIG. 83. Such a coiled column **3050** can be made from wire having a round or rectangular cross-section and can comprise a resiliently flexible material, such as Nitinol, providing lateral flexibility for conforming to the native valve structure while maintaining longitudinal column stiffness for delivery. In some of these embodiments, the frame body **3006** can comprise a quadrhelical coil of four wires having four atrial ends that extend to form the atrial anchors **3012**, **3014** and four ventricular ends that extend to form the four ventricular anchors **3008**, **3010**.

(169) In other embodiments, the frame body **3006** can comprise a plurality of longitudinal members (not shown). In one such example, the frame body **3006** can comprise four longitudinal members: two longitudinal members that extend to form the anterior anchors **3012**, **3014** and two longitudinal members that extend to from the posterior anchors **3008**, **3010**.

(170) In other embodiments, the frame body **3006** can comprise a zig-zag cut pattern **3050** along the longitudinal direction of the body, as shown in FIG. 81, that can also provide lateral flexibility while maintaining column strength.

(171) In some embodiments, the prosthetic spacer **3000** can have additional anchors. In some

embodiment (not shown), the prosthetic spacer **3000** can have three pairs of anchors: one pair of anchors centered around the posterior leaflet **12**, such as the posterior anchors **3010** and **3014** described above, and one pair of anchors at each commissure **36** between the native leaflets **10**, **12**. These commissure anchors pairs can similarly comprise a ventricular anchor and an atrial anchor and can similarly compress the native annulus **8**. In other embodiments, the anterior anchors **3008** and **3012** can also be included as a fourth pair of anchors. Other embodiments can comprise other combinations of these four pairs of anchors and/or additional anchors.

(172) In addition to filling the regurgitant orifice area and blocking blood from flowing toward the left atrium, the prosthetic spacer **3000** can also add tension to the chordae tendinae to prevent further enlargement of the left ventricle and prevent further dilation of the mitral valve annulus.

(173) Anchoring Beneath the Mitral Valve Commissures

(174) Some embodiments of prosthetic devices comprising ventricular anchors, including both prosthetic valves and prosthetic spacers, can be configured such that the ventricular anchors anchor beneath the commissures **36** of the native mitral valve **2** instead of, or in addition to, anchoring behind the A2/P2 regions of the native mitral leaflets **10**, **12**. FIGS. **84-87** show exemplary prosthetic device embodiments that comprise ventricular anchors that anchor beneath the two commissures **36** of the native mitral valve **2**, and related delivery methods. These commissure-anchoring concepts are primarily for use with prosthetic valves, but can be used with other prosthetic devices, including prosthetic spacers.

(175) As shown in FIGS. **3**, **4** and **88**, the commissures **36** are the areas of the native mitral valve **2** where the anterior leaflet **10** and the posterior leaflet **12** are joined. Portions **39** of the native mitral annulus **8** adjacent to each commissure **36**, as shown in FIG. **88**, can be relatively thicker and/or stronger than the portions of the mitral annulus **8** adjacent to the intermediate portions of the leaflets A2/P2, providing a rigid, stable location to anchor a prosthetic apparatus. These annulus regions **39** can comprise tough, fibrous tissue that can take a greater load than the native leaflet tissue, and can form a natural concave surface, or cavity.

(176) FIGS. **84** and **85** show an exemplary prosthetic apparatus **4000** being implanted at the native mitral valve region **2** by positioning a ventricular anchor **4002** at one of the cavities **39**. The prosthetic apparatus **4000** can be a prosthetic valve having a leaflet structure or a spacer device having a spacer body **3004** for reducing MR. The chordae tendinae **16** attach to the leaflets **10**, **12** adjacent to the commissures **36**, which can present an obstacle in positioning ventricular anchors in the cavities **39** behind the chordae. It is possible, however, to deliver anchors, such as anchor **4002**, around the chordae **16** to reach the cavities **39**. Positioning engagement portions, such as the engagement portion **4004**, of the ventricular anchors behind the chordae **16** in these natural cavities **39** can be desirable for retaining a prosthetic apparatus at the native mitral valve region **2**. However, to avoid entanglement with and/or damage to the native chordae **16**, it can be desirable to first guide the engagement portions of the anchors vertically behind the leaflets **10**, **12** at the A2/P2 regions, between the chordae **16** from the postero-medial papillary muscle **22** and the chordae **16** from the antero-lateral papillary muscle **24**, as shown in FIG. **84**, and then move or rotate the engagement portions of the anchors horizontally around behind the chordae **16** toward the commissure cavities **39**, as shown in FIG. **85**.

(177) In some such methods, the ventricular anchors are first deployed behind the A2/P2 regions of the leaflets and then the entire prosthetic apparatus is rotated or twisted to move the engagement portions of the anchors horizontally toward the cavities **39**, as shown in FIGS. **84** and **85**. For example, a first anchor deployed behind the anterior leaflet **10** can move toward one of the cavities **39** while a second anchor deployed behind the posterior leaflet **12** can move toward the other cavity **39**. This method can also be referred to as a “screw method” because the entire prosthetic is rotated to engage the anchors with the native tissue.

(178) As shown in FIGS. **84** and **85**, a prosthetic apparatus **4000** comprising bent, curved, hooked, or generally “L” shaped, anchors **4002** can be used with the screw method. The “L” shaped anchors **4002** can comprise a leg portion **4006** that extends vertically upward from the body of the apparatus **4000**, a knee portion **4008**, and a foot portion **4010** extending horizontally from the knee portion and terminating in the engagement portion **4004**. In some of these embodiments, the “L” shaped anchor

**4002** can comprise a looped wire that attaches to the body of the apparatus **4000** at two locations, such that the wire forms a pair of leg portions **4006**, a pair of knee portions **4008** and a pair of foot portions **4010**. In other embodiments, the anchor **4002** can have other similar shapes, such as a more arced shape, rather than the right angle shape shown in FIG. **84**. During delivery into the heart, the foot portion **4010** can be curled or wrapped around the outer surface of the body of the apparatus **4000**.

(179) As shown in FIG. **84**, in order to move the foot portion **4010** vertically behind the leaflet **10** without entanglement with the chordae, the leg portion **4006** can be positioned slightly off center from the A2 region, closer to the chordae opposite the cavity **39** of desired delivery. As shown in FIG. **84**, the leg portion **4006** is positioned to the right such that the foot portion **4010** can pass between the chordae **16**.

(180) After the foot portion **4010** clears the chordae **16** and is positioned behind the leaflet, the apparatus **4000** can be rotated to move the engagement portion **4004** horizontally into the cavity **39**, as shown in FIG. **85**. Note that in FIG. **85** the leg portion **4006** can end up positioned at the A2/P2 region between the chordae **16** to avoid interference with the chordae.

(181) While FIGS. **84** and **85** show a single anchor **4002**, both an anterior and a posterior anchor can be delivery in symmetrical manners on opposite sides of the native valve **2**, one being anchored at each cavity **39**. The feet **4010** of the two anchors **4002** can point in opposite directions, such that the twisting motion shown in FIG. **85** can move them simultaneously to the two cavities **39**. During delivery of two anchor embodiments, the two foot portions **4010** can wrap around the outer surface of the body of the apparatus **4000** such that the two foot portions **4010** overlap one another.

(182) In similar embodiments, the anchors **4002** can comprise a paddle shape (see FIG. **37** for example) having two foot portions **4010** extending in opposite directions. While more difficult to move between the chordae, these paddle shaped anchors can allow the apparatus **4000** to be rotated in either direction to engage one of the foot portions **4010** at a cavity **39**. In some embodiments, the paddle shaped anchors can be wide enough such that one foot portion **4010** can be positioned at one cavity **39** while the other foot portion is positioned at the other cavity.

(183) Because the anchors **4002** each attach to the body of the apparatus **4000** at two locations, the anchors can spread apart from the main body of the apparatus when the main body is compressed, forming a gap to receive a leaflet, as described in detail above with reference to FIGS. **11-22**. In some embodiments, the anchors can separate from the main body when the main body is compressed and either remain separated from the main body, such that the leaflets are not pinched or compressed between the anchors and the main body of the apparatus, or close against the main body during expansion to engage the leaflets. In some embodiments, the main body can move toward the anchors to reduce the gap when then main body expands while maintaining the distance between the foot portions **4010** of the opposing anchors.

(184) FIGS. **86** and **87** shown another exemplary prosthetic apparatus **5000** being implanted at the native mitral valve region **2** by positioning ventricular anchors **5002** at the cavities **39** and a corresponding method for do so. In this embodiment, the apparatus **5000** can comprise a pair of "L" shaped anchors **5002** on each side (only one pair is visible in FIGS. **86** and **87**), with each pair comprising one anchor for positioning in one of the cavities **39** and another anchor for positioning in the other cavity. Each of the anchors can comprise a leg portion **5006** extending vertically from the body of the apparatus **5000** to a knee portion **5008**, and a foot portion **5010** extending horizontally from the knee portion **5008** to an engagement portion **5004**. In other embodiments, the anchors **5002** can have other similar shapes, such as a more arced shape, rather than the angled shape shown in FIG. **86**.

(185) Each pair of anchors **5002** can comprise a resiliently flexible material, such as Nitinol, such that they can be pre-flexed and constrained in a cocked position for delivery behind the leaflets, as shown in FIG. **86**, and then released to resiliently spring apart to move the engagement portions **5004** in opposite directions toward the two cavities **39**, as shown in FIG. **87**. Any suitable constraint and release mechanisms can be used, such as a releasable mechanical lock mechanism. Once released, one anterior anchor and one posterior anchor can be positioned at one cavity **39** from opposite directions, and a second anterior anchor and a second posterior anchor can be positioned at the other cavity from opposite directions. Some embodiments can include only one anchor on each side of the apparatus **5000** that

move in opposite directions toward opposite cavities **39** when released.

(186) Because each pair of anchors **5002** are initially constrained together, as shown in FIG. **86**, each pair of anchors can act like a single anchor having two attachment points to the main body of the apparatus **5000**. Thus, the anchor pairs can separate, or expand away, from the main body when the main body is compressed and either remain spaced from the main body, such that the leaflets are not pinched or compressed between the anchors and the main body of the apparatus, or close against the main body during expansion to engage the leaflets. In some embodiments, the main body can move toward the anchor pairs to reduce the gap when then main body expands while maintaining the distance between the foot portions **5010** of the opposing anchor pairs.

(187) In the embodiments shown in FIGS. **84-87**, the prosthetic apparatus **4000** or **5000** can have a main frame body similar to the embodiments shown in FIG. **5**, from which the ventricular anchors **4002**, **5002** can extend, and can further comprise one or more atrial anchors, such as an atrial sealing member similar to the atrial sealing member **124** shown in FIG. **5** or a plurality of atrial anchors similar to the atrial anchors **3012** and **3014** shown in FIG. **71**, for example. The atrial anchors can extend radially outward from an atrial end of the prosthetic apparatus and contact the native tissue opposite the cavities **39** and thereby compress the tissue between the atrial anchors and the engagement portions **4004**, **5004** of the ventricular anchors **4002**, **5002** to retain the prosthetic apparatus at the native mitral valve region. The atrial anchors and the ventricular anchors can comprise a broad contact area to distribute the load over a wider area and reduce the likelihood of damaging the native tissue.

(188) In view of the many possible embodiments to which the principles disclosed herein may be applied, it should be recognized that the illustrated embodiments are only preferred examples and should not be taken as limiting the scope of the disclosure. Rather, the scope is defined by the following claims. We therefore claim all that comes within the scope and spirit of these claims.

## Claims

1. A prosthetic heart valve for implantation at a native valve of a heart, the prosthetic heart valve comprising: a multi-part frame comprising: a first annular body comprising a plurality of struts arranged in a lattice pattern and defining a lumen, wherein the first annular body has an inflow end portion and an outflow end portion; and an anchor component comprising a second annular body circumscribing the first annular body and one or more ventricular anchors extending from the second annular body toward the inflow end portion of the first annular body, wherein the one or more ventricular anchors are configured to be positioned behind native valve leaflets; a fabric layer disposed on an inner surface of the first annular body and sutured to the struts; a valve structure comprising three leaflets made from pericardium, wherein the valve structure is disposed within the lumen and inlet ends of the leaflets are sutured to the fabric layer; and wherein the multi-part frame is radially compressible for delivery within a sheath of a delivery catheter and is self-expandable for deployment within an annulus of the native valve.
2. The prosthetic heart valve of claim 1, wherein the multi-part frame further comprises an atrial anchor that extends radially outward from the inflow end portion of the first annular body, wherein the atrial anchor is shaped for placement on an atrial side of the native valve when the prosthetic heart valve is implanted therein.
3. The prosthetic heart valve of claim 2, wherein the atrial anchor extends radially outward from the inflow end portion of the first annular body and then axially upward relative to the inflow end portion of main body.
4. The prosthetic heart valve of claim 2, further comprising a fabric layer extending over the atrial anchor forming an atrial sealing member that is shaped to contact tissue on an atrial side of the native valve when the prosthetic heart valve is implanted therein for forming a seal.
5. The prosthetic heart valve of claim 1, wherein the second annular body comprises a circumferentially extending row of angled struts.
6. The prosthetic heart valve of claim 5, wherein the one or more ventricular anchors comprises a plurality of ventricular anchors and the row of angled struts includes angled struts that are located

circumferentially between locations where the ventricular anchors are connected to the second annular body.

7. The prosthetic heart valve of claim 6, wherein the ventricular anchors are connected to the second annular body at locations between adjacent ends of adjacent angled struts.

8. The prosthetic heart valve of claim 7, wherein the second annular body is connected to the first annular body via a locking mechanism.

9. The prosthetic heart valve of claim 1, wherein the lattice pattern of the first annular body comprises a first row of cells defining an inflow end of the first annular body and a second row of cells defining an outflow end of the first annular body, wherein the valve structure is mounted within the lumen entirely between the inflow end and the outflow end.

10. The prosthetic heart valve of claim 1, wherein the one or more ventricular anchors are covered with fabric.

11. The prosthetic heart valve of claim 1, wherein the second annular body is connected to the first annular body at locations on the first annular body intermediate inflow and outflow ends of the first annular body.

12. A prosthetic heart valve for implantation at a native valve of a heart, the prosthetic heart valve comprising: a multi-part frame comprising: a valve frame comprising: a first annular body comprising a plurality of struts arranged in a lattice pattern and defining a lumen, wherein the first annular body has an inflow end portion and an outflow end portion; an atrial anchor shaped to engage native tissue on an atrial side of the native valve, wherein the atrial anchor extends radially outward from the inflow end portion of the first annular body; and an anchor component comprising: a second annular body; and a plurality of ventricular anchors that extend from the second annular body toward the inflow end portion of the first annular body, wherein the second annular body extends completely around the first annular body, wherein the ventricular anchors are configured to be positioned behind native valve leaflets; a valve structure comprising three leaflets made from pericardium, wherein the valve structure is mounted within the lumen; and a first fabric layer disposed on an inner surface of the first annular body and sutured to the struts, wherein inlet ends of the leaflets are sutured to the first fabric layer along U-shaped seams; wherein the multi-part frame is radially compressible for delivery within a sheath of a delivery catheter and is self-expandable for deployment within an annulus of the native valve.

13. The prosthetic heart valve of claim 12, wherein the second annular body is connected to the first annular body with a locking mechanism.

14. The prosthetic heart valve of claim 12, wherein the second annular body is sewn to the first annular body.

15. The prosthetic heart valve of claim 12, wherein the second annular body comprises a circumferentially extending row of angled struts.

16. The prosthetic heart valve of claim 15, wherein the ventricular anchors are longer than the angled struts.

17. The prosthetic heart valve of claim 12, wherein the valve frame and the anchor component are made of Nitinol.

18. The prosthetic heart valve of claim 12, wherein the ventricular anchors are covered with fabric.

19. The prosthetic heart valve of claim 12, wherein the ventricular anchors comprise six or more ventricular anchors that can engage multiple locations on the native valve leaflets and/or native commissures.

20. A prosthetic heart valve for implantation at a native valve of a heart, the prosthetic heart valve comprising: a first annular frame that is radially expandable from a radially compressed state to a radially expanded state, the first annular frame comprising a plurality of struts arranged in a lattice pattern and defining a lumen, wherein the first annular frame has an inflow end portion and an outflow end portion; an atrial anchor shaped to engage native tissue on an atrial side of the native valve, wherein the atrial anchor extends radially outward from the inflow end portion of the first annular frame and then axially upward relative to the inflow end portion of the first annular frame; a second annular frame that is radially expandable from a radially compressed state to a radially expanded state, wherein the second annular frame comprises a circumferentially extending row of angled struts that extends around

the first annular frame; a plurality of ventricular anchors that extend from the second annular frame toward the inflow end portion of the first annular frame, wherein the ventricular anchors are configured to be positioned behind native valve leaflets; a valve structure comprising three leaflets made from pericardium, wherein the valve structure is mounted within the lumen; and a fabric layer disposed on an inner surface of the first annular frame and sutured to the struts of the first annular frame, wherein inlet ends of the leaflets are sutured to the fabric layer along U-shaped seams.

21. The prosthetic heart valve of claim 20, wherein the fabric layer extends over the atrial anchor.

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